

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:14:37

Source: data/json/noise_experiments/v2/noise2i_match8_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_match8/noise2i_match8_game_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_match8
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 4
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$6; payoffs Agent_1=\$8, Agent_2=\$3;

noise/visible: sent visible=\$3.95, received visible=\$11.85, returned visible=\$5.02

Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 8.0,

"Agent_2": 3.0

}

Visible balances after round: {

"Agent_1": 7.02,

"Agent_2": 5.8500000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

Reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent /returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)
- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)

- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05 (a \$2.05 gain)
- I end up with: $\$11.85 - \$6 = \$5.85$ (a \$5.85 gain)

This rewards their trust with a good return while maintaining mutual benefit.

```
{"return": 6.00}
```

Reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

The sender was quite generous, sending nearly 80% of their endowment. This shows significant trust.

Let me think about what would be fair:

- The sender gave up \$3.95 and kept \$1.05
- I received \$11.85
- If I want to be reciprocal and fair, I should return enough so that:
 - The sender ends up with more than they started with (to reward their trust)
 - I also benefit from the exchange

Let me consider different return amounts:

If I return \$6:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05
- I end up with: $\$11.85 - \$6 = \$5.85$
- Both better off, sender gains \$2.05, I gain \$5.85

If I return \$5:

- Sender ends up with: $\$5 - \$3.95 + \$5 = \6.05
- I end up with: $\$11.85 - \$5 = \$6.85$
- Sender gains \$1.05, I gain \$6.85

If I return \$7:

- Sender ends up with: $\$5 - \$3.95 + \$7 = \8.05
- I end up with: $\$11.85 - \$7 = \$4.85$
- Sender gains \$3.05, I gain \$4.85

A common approach in trust games is to split the gains somewhat fairly.

The sender's trust created \$7.90 in additional value beyond what they sent. An equal split of the total pool would give each of us \$6.45, meaning I'd return \$5.40. However, since the sender took the initial risk, returning \$6 seems more appropriate-it rewards their trust while still leaving me with a meaningful gain....

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 844,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$0; payoffs Agent_2=\$1, Agent_1=\$12;
 noise/visible: sent visible=\$0, received visible=\$0
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 12.0, "min": 0, "raw":

```

    0.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 20.0,
  "Agent_2": 4.0
}
Visible balances after round: {
  "Agent_1": 7.02,
  "Agent_2": 6.850000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:28
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)
- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05 (a \$2.05 gain)
- I end up with: $\$11.85 - \$6 = \$5.85$ (a \$5.85 gain)

This rewards their trust with a good return while maintaining mutual benefit.

{"return": 6.00}

reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

The sender was quite generous, sending nearly 80% of their endowment. This shows significant trust.

Let me think about what would be fair:

- The sender gave up \$3.95 and kept \$1.05
- I received \$11.85
- If I want to be reciprocal and fair, I should return enough so that:
 - The sender ends up with more than they started with (to reward their trust)
 - I also benefit from the exchange

Let me consider different return amounts:

If I return \$6:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05
- I end up with: $\$11.85 - \$6 = \$5.85$
- Both better off, sender gains \$2.05, I gain \$5.85

If I return \$5:

- Sender ends up with: $\$5 - \$3.95 + \$5 = \6.05
- I end up with: $\$11.85 - \$5 = \$6.85$
- Sender gains \$1.05, I gain \$6.85

If I return \$7:

- Sender ends up with: $\$5 - \$3.95 + \$7 = \8.05
- I end up with: $\$11.85 - \$7 = \$4.85$
- Sender gains \$3.05, I gain \$4.85

A common approach in trust games is to split the gains somewhat fairly.

The sender's trust created \$7.90 in additional value beyond what they sent. An equal split of the total pool would give each of us \$6.45, meaning I'd return \$5.40. However, since the sender took the initial risk, returning \$6 seems more appropriate-it rewards their trust while still leaving me with a meaningful gain....

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 844,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.850000000000000

1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience from Round 1:

- The previous sender showed strong trust by sending 79% of their endowment
- I reciprocated by returning about 50% of what I received
- This created mutual benefit for both of us

I want to maintain this cooperative dynamic. Since trust was shown to me and rewarded, I should demonstrate similar trust as the sender.

I'll send \$4.00 (80% of my endowment), which mirrors the high level of trust shown to me in Round 1. This signals my willingness to cooperate and hopefully encourages reciprocity.

```
{"send": 4.00}
```

Reasoning:

Now I'm the SENDER in round 2.

Looking at my previous experience:

- In Round 1, I was the receiver
- The sender sent me \$3.95 (79% of their endowment)
- I received \$11.85 and returned \$6.00
- This worked out well for both of us - they gained \$2.05 and I gained \$5.85

Now I'm in the sender position. I should consider:

1. The previous sender showed high trust (79%)
2. I reciprocated fairly by returning about 50% of what I received
3. This created mutual benefit

Since the previous sender trusted me with 79% of their endowment and it worked out well, I should also show substantial trust. This could encourage cooperation and build a good reciprocal relationship.

Given that there may be communication noise, I should send a clear, substantial amount that demonstrates trust.

I'll send around \$4.00 (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around \$6, ending up with \$7 total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

Usage: {

```
"input_tokens": 634,
"output_tokens": 460,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:10:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
```

```

"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)
- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER

- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

Reasoning:

Let me understand this situation:

- Round 1: I was the SENDER and sent \$3
- I started with \$5
 - I sent \$3 (so the receiver got \$9)
 - My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

```
Usage: {
  "input_tokens": 593,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

```
dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $0; payoffs Agent_1=$2, Agent_2=$9;
noise/visible: sent visible=$0, received visible=$0, returned visible=$0.81
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw":
0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 22.0,
  "Agent_2": 13.0
}
Visible balances after round: {
  "Agent_1": 9.83,
  "Agent_2": 6.850000000000001
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:10:57
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)

- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json
{'send': 3}
'''
```

Reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total earnings show \$7.02

If I kept \$2 from Round 1, and now have total earnings of \$7.02, that means I received back \$5.02 in Round 1.

That's a very generous return! The receiver returned \$5.02 out of the \$9 they received.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I returned \$0
- So Round 2 added \$0 to my earnings

My total is still \$7.02, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent \$3, the receiver was very generous and returned \$5.02 out of \$9 (keeping only \$3.98)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify and cooperate with them.

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

```
Usage: {
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:11:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)
- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: \$5 - \$3.95 + \$6 = \$7.05 (a \$2.05 gain)
- I end up with: \$11.85 - \$6 = \$5.85 (a \$5.85 gain)

This rewards their trust with a good return while maintaining mutual benefit.

{"return": 6.00}

reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

The sender was quite generous, sending nearly 80% of their endowment. This shows significant trust.

Let me think about what would be fair:

- The sender gave up \$3.95 and kept \$1.05
- I received \$11.85
- If I want to be reciprocal and fair, I should return enough so that:
 - The sender ends up with more than they started with (to reward their trust)
 - I also benefit from the exchange

Let me consider different return amounts:

If I return \$6:

- Sender ends up with: \$5 - \$3.95 + \$6 = \$7.05
- I end up with: \$11.85 - \$6 = \$5.85
- Both better off, sender gains \$2.05, I gain \$5.85

- Sender ends up with: $\$5 - \$3.95 + \$5 = \6.05
- I end up with: $\$11.85 - \$5 = \$6.85$
- Sender gains \$1.05, I gain \$6.85

- Sender ends up with: $\$5 - \$3.95 + \$7 = \8.05
- I end up with: $\$11.85 - \$7 = \$4.85$
- Sender gains $\$3.05$, I gain $\$4.85$

The sender's trust created \$7.90 in additional value beyond what they sent. An equal split of the total pool would give each of us \$6.45, meaning I'd return \$5.40. However, since the sender took the initial risk, returning \$6 seems more appropriate-it rewards their trust while still leaving me with a meaningful gain....

```
{
  "input_tokens": 402,
  "output_tokens": 844,
  "reasoning_tokens": 0
}
```

How much do you send to the receiver this round? (0-5)

- The previous sender showed strong trust by sending 79% of their endowment
- I reciprocated by returning about 50% of what I received
- This created mutual benefit for both of us

I'll send around \$4.00 (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around \$6, ending up with \$7 total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

usage:

```
{
  "input_tokens": 634,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.8500000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds. I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

```
{"return": 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds where there was high trust and cooperation.

Since I received \$0, I have nothing to return - the maximum I can return is \$0 (as stated in the prompt).

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

Either way, I cannot return anything because I received nothing.

My total earnings remain at \$6.85 from previous rounds.

Usage: {

```
"input_tokens": 838,
"output_tokens": 207,
"reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.5; received \$7.5; Agent_1 returned \$0; payoffs Agent_2=\$2.5, Agent_1=\$7.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.92

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

```
"Agent_1": 29.5,
"Agent_2": 15.5
}
```

Visible balances after round: {

```
"Agent_1": 9.83,
"Agent_2": 10.270000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:11:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

content:

Looking at my experience from Round 1:

- The previous sender showed strong trust by sending 79% of their endowment
- I reciprocated by returning about 50% of what I received
- This created mutual benefit for both of us

I want to maintain this cooperative dynamic. Since trust was shown to me and rewarded, I should demonstrate similar trust as the sender.

I'll send \$4.00 (80% of my endowment), which mirrors the high level of trust shown to me in Round 1. This signals my willingness to cooperate and hopefully encourages reciprocity.

```
{"send": 4.00}
```

reasoning:

Now I'm the SENDER in round 2.

Looking at my previous experience:

- In Round 1, I was the receiver
- The sender sent me \$3.95 (79% of their endowment)
- I received \$11.85 and returned \$6.00
- This worked out well for both of us - they gained \$2.05 and I gained \$5.85

Now I'm in the sender position. I should consider:

1. The previous sender showed high trust (79%)
2. I reciprocated fairly by returning about 50% of what I received
3. This created mutual benefit

Since the previous sender trusted me with 79% of their endowment and it worked out well, I should also show substantial trust. This could encourage cooperation and build a good reciprocal relationship.

Given that there may be communication noise, I should send a clear, substantial amount that demonstrates trust.

I'll send around \$4.00 (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around \$6, ending up with \$7 total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

usage:

```
{
  "input_tokens": 634,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.850000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant**content:**

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds. I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds where there was high trust and cooperation.

Since I received \$0, I have nothing to return - the maximum I can return is \$0 (as stated in the prompt).

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

Either way, I cannot return anything because I received nothing.

My total earnings remain at \$6.85 from previous rounds.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.8500000000000000

1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 1: As receiver, someone trusted me with 79%, I returned fairly
- Round 2: I trusted the receiver with 80%, but received nothing back (lost \$4)
- Round 3: As receiver, I got nothing sent to me

The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I want to signal willingness to cooperate while being cautious given recent outcomes. Sending nothing perpetuates the breakdown, but sending too much risks another significant loss.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

```
{"send": 2.50}
```

Reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00
Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...
Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

My total earnings are \$6.85.

Let me trace through more carefully:

- Round 1 ended: I had \$5.85 (started with \$0, received \$11.85, returned \$6.00)
- Round 2: I sent \$4.00 as sender, so I had \$1.00 left plus whatever was returned to me
 - If my total is now \$6.85, that means in Round 2, I ended with \$6.85 - \$5.85 = \$1.00
 - I started Round 2 with \$5, sent \$4.00, kept \$1.00
 - I received back \$0 (since \$1.00 + \$0 = \$1.00)
 - So the receiver in Round 2 got \$12.00 (my \$4 × 3) and returned \$0!
- Round 3: I was receiver, got \$0, so my earnings stayed at \$6.85

So what happened:

- Round 1: Sender trusted me (79%), I reciprocated fairly (returned ~50%)
- Round 2: I trusted the receiver (80%), they betrayed me completely (returned 0%)
- Round 3: New sender didn't trust me at all (sent 0%)

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

```
Usage: {  
  "input_tokens": 952,  
  "output_tokens": 910,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:11:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)

- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json
{'send': 3}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total earnings show \$7.02

If I kept \$2 from Round 1, and now have total earnings of \$7.02, that means I received back \$5.02 in Round 1.

That's a very generous return! The receiver returned \$5.02 out of the \$9 they received.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I returned \$0
- So Round 2 added \$0 to my earnings

My total is still \$7.02, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent \$3, the receiver was very generous and returned \$5.02 out of \$9 (keeping only \$3.98)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify and cooperate with them.

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Round 4 Status:

- I'm the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, there are no funds available to return. The only possible response is:

```
'''json
{'return': 0}
'''
```

Reflection: This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

Reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.

After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

```
Usage: {
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;
noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.27
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 32.5,
"Agent_2": 21.5

}

Visible balances after round: {

"Agent_1": 13.1,
"Agent_2": 10.270000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:11:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},

"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish

- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)
- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json
{'send': 3}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total earnings show $\$7.02$

If I kept $\$2$ from Round 1, and now have total earnings of $\$7.02$, that means I received back $\$5.02$ in Round 1.
That's a very generous return! The receiver returned $\$5.02$ out of the $\$9$ they received.

Round 2: I was the RECEIVER

- The sender sent me $\$0$
- I received $\$0 \times 3 = \0
- I returned $\$0$
- So Round 2 added $\$0$ to my earnings

My total is still $\$7.02$, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent $\$3$, the receiver was very generous and returned $\$5.02$ out of $\$9$ (keeping only $\$3.98$)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify and cooperate with them.

I'll send $\$3$ again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are $\$9.83$.

This round, The sender sent you $\$0$, so you received $\$0$. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

```
**Round 4 Status:**
- I'm the RECEIVER
- The sender sent me $0
- After the 3x multiplier:  $\$0 \times 3 = \$0$ 
- I received nothing
```

Since I received $\$0$, there are no funds available to return. The only possible response is :

```
'''json
{'return': 0}
'''
```

Reflection: This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.
After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.

How much do you send to the receiver this round? (0-5)

Assistant output:

****Round 5 Analysis:****

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return
- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate
- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

Reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: $\$2 + \$5.02 = \$7.02$
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: $\$2 + \$0.81 = \$2.81$
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

...

```
Usage: {
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:12:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)
- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: \$5 - \$3.95 + \$6 = \$7.05 (a \$2.05 gain)
- I end up with: \$11.85 - \$6 = \$5.85 (a \$5.85 gain)

This rewards their trust with a good return while maintaining mutual benefit.

{"return": 6.00}

reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

The sender was quite generous, sending nearly 80% of their endowment. This shows significant trust.

Let me think about what would be fair:

- The sender gave up \$3.95 and kept \$1.05
- I received \$11.85
- If I want to be reciprocal and fair, I should return enough so that:
 - The sender ends up with more than they started with (to reward their trust)
 - I also benefit from the exchange

Let me consider different return amounts:

If I return \$6:

- Sender ends up with: \$5 - \$3.95 + \$6 = \$7.05
- I end up with: \$11.85 - \$6 = \$5.85
- Both better off, sender gains \$2.05, I gain \$5.85

If I return \$5:

- Sender ends up with: \$5 - \$3.95 + \$5 = \$6.05
- I end up with: \$11.85 - \$5 = \$6.85
- Sender gains \$1.05, I gain \$6.85

If I return \$7:

- Sender ends up with: $\$5 - \$3.95 + \$7 = \8.05
- I end up with: $\$11.85 - \$7 = \$4.85$
- Sender gains $\$3.05$, I gain $\$4.85$

A common approach in trust games is to split the gains somewhat fairly.

The sender's trust created $\$7.90$ in additional value beyond what they sent. An equal split of the total pool would give each of us $\$6.45$, meaning I'd return $\$5.40$. However, since the sender took the initial risk, returning $\$6$ seems more appropriate-it rewards their trust while still leaving me with a meaningful gain....

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 844,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$5.8500000000000000$ 1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience from Round 1:

- The previous sender showed strong trust by sending 79% of their endowment
- I reciprocated by returning about 50% of what I received
- This created mutual benefit for both of us

I want to maintain this cooperative dynamic. Since trust was shown to me and rewarded, I should demonstrate similar trust as the sender.

I'll send $\$4.00$ (80% of my endowment), which mirrors the high level of trust shown to me in Round 1. This signals my willingness to cooperate and hopefully encourages reciprocity.

```
{"send": 4.00}
```

reasoning:

Now I'm the SENDER in round 2.

Looking at my previous experience:

- In Round 1, I was the receiver
- The sender sent me $\$3.95$ (79% of their endowment)
- I received $\$11.85$ and returned $\$6.00$
- This worked out well for both of us - they gained $\$2.05$ and I gained $\$5.85$

Now I'm in the sender position. I should consider:

1. The previous sender showed high trust (79%)
2. I reciprocated fairly by returning about 50% of what I received
3. This created mutual benefit

Since the previous sender trusted me with 79% of their endowment and it worked out well, I should also show substantial trust. This could encourage cooperation and build a good reciprocal relationship.

Given that there may be communication noise, I should send a clear, substantial amount that demonstrates trust.

I'll send around $\$4.00$ (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around $\$6$, ending up with $\$7$ total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

usage:

```
{
  "input_tokens": 634,
```

Message 6 role=user

Round 3

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

content:

This is quite different from the cooperative pattern we had established in earlier rounds. I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

reasoning:

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

My total earnings remain at \$6.85 from previous rounds.

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

content:

Round 4

How much do you send to the receiver this round? (0-5)

content:

- Round 1: As receiver, someone trusted me with 79%, I returned fairly
- Round 2: I trusted the receiver with 80%, but received nothing back (lost \$4)
- Round 3: As receiver, I got nothing sent to me

The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I want to signal willingness to cooperate while being cautious given recent outcomes. Sending nothing perpetuates the breakdown, but sending too much risks another significant loss.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00

Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...

Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

My total earnings are \$6.85.

Let me trace through more carefully:

- Round 1 ended: I had \$5.85 (started with \$0, received \$11.85, returned \$6.00)
- Round 2: I sent \$4.00 as sender, so I had \$1.00 left plus whatever was returned to me
 - If my total is now \$6.85, that means in Round 2, I ended with \$6.85 - \$5.85 = \$1.00
 - I started Round 2 with \$5, sent \$4.00, kept \$1.00
 - I received back \$0 (since \$1.00 + \$0 = \$1.00)
 - So the receiver in Round 2 got \$12.00 (my \$4 × 3) and returned \$0!
- Round 3: I was receiver, got \$0, so my earnings stayed at \$6.85

So what happened:

- Round 1: Sender trusted me (79%), I reciprocated fairly (returned ~50%)
- Round 2: I trusted the receiver (80%), they betrayed me completely (returned 0%)
- Round 3: New sender didn't trust me at all (sent 0%)

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 910,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.270000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

```
{"return": 0}
```

Reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85

- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
- So I must have received $5.85 - 1 = 0.85$? That seems low. Let me think...
- I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
- If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2
- In R2: $5 - 4 + \text{return} = \text{gain}$
- So $1 + \text{return} = \text{gain}$
- And $5.85 + \text{gain} = 6.85$
- So $\text{gain} = 1$
- So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

```
Usage: {
  "input_tokens": 1196,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 sent $0; received $0; Agent_1 returned $0; payoffs Agent_2=$5, Agent_1=$0;
noise/visible: sent visible=$0.05, received visible=$0.15
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.1},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
"Agent_1": 32.5,
"Agent_2": 26.5
}
```

Visible balances after round: {

```
"Agent_1": 13.25,
"Agent_2": 15.270000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:12:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)
- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: \$5 - \$3.95 + \$6 = \$7.05 (a \$2.05 gain)
- I end up with: \$11.85 - \$6 = \$5.85 (a \$5.85 gain)

This rewards their trust with a good return while maintaining mutual benefit.

{"return": 6.00}

reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

- This worked out well for both of us - they gained \$2.05 and I gained \$5.85

Now I'm in the sender position. I should consider:

1. The previous sender showed high trust (79%)
2. I reciprocated fairly by returning about 50% of what I received
3. This created mutual benefit

Since the previous sender trusted me with 79% of their endowment and it worked out well, I should also show substantial trust. This could encourage cooperation and build a good reciprocal relationship.

Given that there may be communication noise, I should send a clear, substantial amount that demonstrates trust.

I'll send around \$4.00 (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around \$6, ending up with \$7 total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

usage:

```
{
  "input_tokens": 634,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.850000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds. I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds where there was high trust and cooperation.

Since I received \$0, I have nothing to return - the maximum I can return is \$0 (as stated in the prompt).

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

Either way, I cannot return anything because I received nothing.

My total earnings remain at \$6.85 from previous rounds.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

How much do you send to the receiver this round? (0-5)

Looking at the pattern:

- The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

reasoning:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00
Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...
Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

Let me trace through more carefully:

- So what happened:

- Now I'm the sender again in Round 4. The pattern seems to have broken down:

- Two considerations:

- I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

```
{
  "input_tokens": 952,
  "output_tokens": 910,
  "reasoning_tokens": 0
}
```

content:

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You are the RECEIVER this round. Your total earnings across all rounds are \$10.270000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

{"return": 0}

reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85
- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
- So I must have received $5.85 - 1 = 0.85$? That seems low. Let me think...
- I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
- If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2
- In R2: $5 - 4 + \text{return} = \text{gain}$
- So $1 + \text{return} = \text{gain}$
- And $5.85 + \text{gain} = 6.85$
- So $\text{gain} = 1$
- So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.270000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

- Round 2: I sent \$4 as sender, received very little back (lost money)
- Round 3: As receiver, got nothing sent to me
- Round 4: I sent \$2.50 as sender, received very little back (lost money again)
- Round 5: As receiver, got nothing sent to me again

The cooperation that existed in Round 1 has completely disappeared. Every time I've sent money as the sender, I've lost money. Every recent round as receiver, I've received nothing.

While the communication noise mentioned in the instructions is a consideration, I have to make decisions based on what I observe. The pattern is clear and consistent: sending money results in losses.

I need to protect my remaining earnings at this point.

```
{"send": 0}
```

Reasoning:

Let me review what has happened:

Round 1: I was receiver, got \$11.85, returned \$6. The sender had sent \$3.95.

Round 2: I was sender, sent \$4, but the next round my earnings were only \$6.85, which means I got \$1 back (I had \$5 - \$4 + return = earnings - previous earnings = \$6.85 - \$5.85 = \$1). So I sent \$4, got \$1 back.

Round 3: I was receiver, got \$0. So the sender sent \$0.

Round 4: I was sender, sent \$2.50. My earnings went from \$6.85 to \$10.27, an increase of \$3.42. So I kept \$5 - \$2.50 + return = \$2.50 + return, and the increase was \$3.42, so return = \$3.42 - \$2.50 = \$0.92. Hmm, wait, let me recalculate. After round 3, I had \$6.85. In round 4 as sender, I start with \$5, send \$2.50, leaving me \$2.50. The return I got was such that my total earnings increased by \$10.27 - \$6.85 = \$3.42. So I earned \$2.50 (what I kept) + return, and this equals \$3.42, so return = \$0.92.

Round 5: I was receiver, got \$0.

So the pattern is:

- Round 1: High cooperation (someone sent me a lot, I returned fairly)
- Round 2: I sent \$4, got only \$1 back (poor return)
- Round 3: I got nothing sent to me
- Round 4: I sent \$2.50, got \$0.92 back (very poor return)
- Round 5: Again received nothing

The other player has consistently failed to reciprocate. My attempts to send money have resulted in substantial losses-\$3 net in round 2 and \$1.58 in round 4. The mention of communication noise raises a possibility: maybe what I intended to send didn't reach them, or their returns didn't reach me. But regardless, I'm facing round 6 as sender with a clear pattern of minimal cooperation. Sending \$0 protects my current earnings but forfeits any potential mutual gain.

There's another angle worth considering-if I'm paired with different people each round, then this person's actions shouldn't be judged by previous rounds' outcomes. The instructions suggest varying roles across rounds, which implies varying partners too. That would make each interaction somewhat fresh rather than a continuation of the same broken dynamic.

Looking at the pattern either way though: whether it's one person or five, I'm seeing consistent poor returns when I cooperate. Four of five rounds show weak reciprocation. The risk-reward calculation keeps pointing toward minimal sending. A \$1 bet might be the compromise-it caps my downside while leaving room for a positive surprise....

```
Usage: {  
  "input_tokens": 1304,  
  "output_tokens": 1383,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:12:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send ****\$3****, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```json

{'send': 3}

```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)
- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json
{'send': 3}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total earnings show \$7.02

If I kept \$2 from Round 1, and now have total earnings of \$7.02, that means I received back \$5.02 in Round 1.

That's a very generous return! The receiver returned \$5.02 out of the \$9 they received.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I returned \$0
- So Round 2 added \$0 to my earnings

My total is still \$7.02, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent \$3, the receiver was very generous and returned \$5.02 out of \$9 (keeping only \$3.98)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify and cooperate with them.

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

****Round 4 Status:****
- I'm the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, there are no funds available to return. The only possible response is :

```
'''json
{'return': 0}
'''
```

****Reflection:**** This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.
After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.
How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

****Round 5 Analysis:****

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return
- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate

- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: $\$2 + \$5.02 = \$7.02$
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: $\$2 + \$0.81 = \$2.81$
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

...

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.1.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Assistant output:

****Round 6 Status:****

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: \$0.15 received

****My decision:****

The sender sent a very small amount, but they still showed some trust by sending something rather than nothing. Given that I've received \$0 in most rounds as receiver, even a small gesture of cooperation is notable.

To signal that I'm trustworthy and encourage future cooperation, I'll return a fair portion that shares the gains from the multiplier:

- If I return \$0.10, the sender nets \$0.05 profit overall
- I keep \$0.05
- This demonstrates I'm willing to cooperate even with small amounts

```
'''json
{'return': 0.10}
'''
```

This way, both of us benefit from the interaction, and I establish myself as a reliable partner for potential future rounds.

Reasoning:

Round 6 Analysis:

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: $\$0.05 \times 3 = \0.15
- My total earnings are \$13.1

This is interesting - the sender sent a very small amount (\$0.05). This could be:

1. A test to see if I'm trustworthy with small amounts
2. The sender being very cautious after bad experiences
3. Communication noise (though \$0.05 seems deliberate)

Given that:

- The amount is tiny (\$0.15 received)
- The sender showed some trust, even if minimal
- I've been getting mostly \$0 as a receiver, so any trust gesture is notable
- I should establish myself as trustworthy if I want future cooperation

I think I should return a fair amount or even slightly more than "fair" to signal trustworthiness. A typical fair split might be 50-50, so I'd return \$0.075 and keep \$0.075.

But given the pattern of non-cooperation, maybe I should return slightly more to build trust - perhaps return \$0.10 and keep \$0.05? This would show I'm cooperative even with small amounts.

Actually, let me think about this differently. The sender:

- Kept \$4.95
- Gave me \$0.05 which became \$0.15
- If I return \$0.10, they end up with $\$4.95 + \$0.10 = \$5.05$ (gained \$0.05)
- I end up with \$0.05

This seems reasonable and fair - I'm sharing the gains from the multiplier. Let me go with returning \$0.10.

```
Usage: {
  "input_tokens": 1556,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0.1; payoffs Agent_1=\$3.1, Agent_2=\$5.9;
noise/visible: sent visible=\$0.05, received visible=\$0.15, returned visible=\$1.03
Action validation: {"returned": {"amount": 0.1, "clamped": false, "max": 6.0, "min": 0, "raw": 0.1}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 35.6,
"Agent_2": 32.4

}

Visible balances after round: {

"Agent_1": 17.28,
"Agent_2": 15.320000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:13:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

****My reasoning:****

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing trust could benefit future interactions
- Sending everything (\$5) is too risky with an unknown partner
- Sending nothing (\$0) misses the opportunity for cooperation
- A moderate amount shows willingness to trust while limiting downside risk

I'll send **\$3**, which:

- Gives the receiver \$9 (tripled)
- Leaves me with \$2 as security
- Signals cooperative intent
- Allows room for fair reciprocation

```
'''json
{'send': 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5 to potentially send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value. This is the cooperative aspect.
2. The receiver then decides how much to return to me.
3. There's a note about "communication noise" - amounts may differ from what was actually sent/returned.

In a trust game, the optimal cooperative strategy is typically to send money because:

- Sending \$5 ? Receiver gets \$15
- If receiver is fair/reciprocal, they might return around \$7.50 or more
- This would give me \$7.50 (more than my initial \$5)
- And receiver keeps \$7.50 (more than \$0)

However, there's risk:

- Receiver could return nothing
- I'd end up with \$0

First round considerations:

- I have no reputation yet to establish
- The receiver doesn't know if I'll be trustworthy in future rounds (if roles reverse)
- Moderate trust might be reasonable to signal cooperation

Common strategies in trust games:

- Send about half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize potential gains but risk losing everything
- Send nothing (\$0) to avoid risk but miss cooperative gains

Given this is round 1 and there's mention of multiple rounds where we might switch roles, showing some trust could establish a cooperative pattern. I'll send a moderate amount - let's say \$3. This:

- Shows willingness to trust
- Creates value (\$9 for receiver)
- Limits my downside risk (I keep \$2)
- Allows receiver to return a fair amount

Actually, let me reconsider. In behavioral experiments, sending around \$2.50-\$3 is common in first rounds. I'll go with \$3.

Wait, I should also consider that there's "communication noise" mentioned. This might mean the receiver sees a different amount than what I send. ...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 658,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
```

```
"input_tokens": 593,  
"output_tokens": 656,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json  
{ 'send': 3 }  
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- My total earnings show \$7.02

If I kept \$2 from Round 1, and now have total earnings of \$7.02, that means I received back \$5.02 in Round 1.

That's a very generous return! The receiver returned \$5.02 out of the \$9 they received.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I returned \$0
- So Round 2 added \$0 to my earnings

My total is still \$7.02, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent \$3, the receiver was very generous and returned \$5.02 out of \$9 (keeping only \$3.98)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify

y and cooperate with them.

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

```
**Round 4 Status:**
- I'm the RECEIVER
- The sender sent me $0
- After the 3x multiplier:  $\$0 \times 3 = \$0$ 
- I received nothing
```

Since I received \$0, there are no funds available to return. The only possible response is :

```
'''json
{'return': 0}
'''
```

****Reflection:**** This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.

After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

```
**Round 5 Analysis:**
```

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return
- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate
- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: \$2 + \$5.02 = \$7.02
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: \$2 + \$0.81 = \$2.81
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

...

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.1.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 13 role=assistant

content:

****Round 6 Status:****

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: \$0.15 received

****My decision:****

The sender sent a very small amount, but they still showed some trust by sending something rather than nothing. Given that I've received \$0 in most rounds as receiver, even a small gesture of cooperation is notable.

To signal that I'm trustworthy and encourage future cooperation, I'll return a fair portion that shares the gains from the multiplier:

- If I return \$0.10, the sender nets \$0.05 profit overall
- I keep \$0.05
- This demonstrates I'm willing to cooperate even with small amounts

```
'''json
{'return': 0.10}
'''
```

This way, both of us benefit from the interaction, and I establish myself as a reliable partner for potential future rounds.

reasoning:

Round 6 Analysis:

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: $\$0.05 \times 3 = \0.15
- My total earnings are \$13.1

This is interesting - the sender sent a very small amount (\$0.05). This could be:

1. A test to see if I'm trustworthy with small amounts
2. The sender being very cautious after bad experiences
3. Communication noise (though \$0.05 seems deliberate)

Given that:

- The amount is tiny (\$0.15 received)
- The sender showed some trust, even if minimal
- I've been getting mostly \$0 as a receiver, so any trust gesture is notable
- I should establish myself as trustworthy if I want future cooperation

I think I should return a fair amount or even slightly more than "fair" to signal trustworthiness. A typical fair split might be 50-50, so I'd return \$0.075 and keep \$0.075.

But given the pattern of non-cooperation, maybe I should return slightly more to build trust - perhaps return \$0.10 and keep \$0.05? This would show I'm cooperative even with small amounts.

Actually, let me think about this differently. The sender:

- Kept \$4.95
- Gave me \$0.05 which became \$0.15
- If I return \$0.10, they end up with $\$4.95 + \$0.10 = \$5.05$ (gained \$0.05)
- I end up with \$0.05

This seems reasonable and fair - I'm sharing the gains from the multiplier. Let me go with returning \$0.10.

usage:

```
{
  "input_tokens": 1556,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.25. How much do you send to the receiver this round? (0-5)

Assistant output:

****Round 7 Analysis:****

Looking at my recent experience:

- Round 5 (Sender): Sent \$2 ? received minimal return
- Round 6 (Receiver): Got only \$0.05, but I returned \$0.10 generously to encourage cooperation

****Current situation:****

My earnings have been growing, but cooperation as sender has been hit-or-miss. However, in Round 6 I demonstrated that I'm a trustworthy receiver by returning more than I kept even from a tiny amount.

****My decision:****

I'll continue testing for cooperative partners. The 3x multiplier creates real potential for mutual benefit. I'll send a moderate amount:

I'll send ****\$2****:

- Shows willingness to build trust
- Creates \$6 for receiver (good potential for both of us)
- Keeps \$3 as security
- Balances cooperation with risk management

```
'''json
{'send': 2}
'''
```

If more receivers prove trustworthy, I might increase; if cooperation continues to fail, I'll be more conservative.

Reasoning:

Let me track my earnings across rounds:

Round 1 (Sender): Sent \$3, kept \$2, got generous return ? ended around \$7.02
Round 2 (Receiver): Got \$0, returned \$0 ? still \$7.02
Round 3 (Sender): Sent \$3, kept \$2, got minimal return (~\$0.81) ? ended around \$9.83
Round 4 (Receiver): Got \$0, returned \$0 ? still \$9.83
Round 5 (Sender): Sent \$2, kept \$3, got return of ~\$0.27 ? ended around \$13.10
Round 6 (Receiver): Got \$0.15, returned \$0.10, kept \$0.05 ? ended around \$13.15
Now Round 7: Starting at \$13.25 (gained \$0.10 from Round 5 return it seems)

Wait, let me recalculate:

- Round 5 total was \$9.83
- Round 6 started at \$13.1, which is \$3.27 higher
- This suggests in Round 5, I sent \$2, kept \$3, and got back around \$3.27? That's a very good return!
- Round 6: I was receiver, got \$0.15, returned \$0.10, kept \$0.05 ? $\$13.1 + \$0.05 = \$13.15$
- Current earnings: \$13.25, so I gained \$0.10 from being sender in some recent round

Actually, looking at the pattern:

- Round 4 ended at \$9.83
- Round 5 I was sender and my earnings should have been: $\$9.83 - \text{what I sent} + \text{what was returned} + \5 new
- Round 6 started at \$13.1

If I sent \$2 in Round 5:

- $\$9.83 - \$2 + \text{return} + \$5 = \13.1
- $\text{return} = \$13.1 - \$9.83 - \$5 + \$2 = \$0.27$

Then Round 6: $\$13.1 + \$0.05 \text{ (kept from } \$0.15) = \$13.15$

Round 7 start: \$13.25

There's a \$0.10 gap I haven't fully accounted for-possibly a small return from an earlier round. The clearer pattern emerging is that when I send money, returns vary dramatically: the first round was generous, the third minimal, and the fifth modest. As a receiver, I've gotten almost nothing back....

```
Usage: {
  "input_tokens": 1803,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:13:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.95, that is 79.0% of their total endowment. You received \$11.850000000000001. How much do you return to The sender? (0-11.850000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.95 (79% of their endowment)
- This created \$11.85 for me to work with
- The sender currently has only \$1.05 remaining

I want to reciprocate their trust and ensure they benefit from their generous decision, while also benefiting myself from this exchange.

If I return \$6.00:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05 (a $\$2.05$ gain)
- I end up with: $\$11.85 - \$6 = \$5.85$ (a $\$5.85$ gain)

This rewards their trust with a good return while maintaining mutual benefit.

```
{"return": 6.00}
```

reasoning:

Let me understand the situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$3.95 (which is 79% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$11.85 ($3.95 \times 3 = 11.85$)
4. I need to decide how much to return (between \$0 and \$11.85)

The sender was quite generous, sending nearly 80% of their endowment. This shows significant trust.

Let me think about what would be fair:

- The sender gave up \$3.95 and kept \$1.05
- I received \$11.85
- If I want to be reciprocal and fair, I should return enough so that:
 - The sender ends up with more than they started with (to reward their trust)
 - I also benefit from the exchange

Let me consider different return amounts:

If I return \$6:

- Sender ends up with: $\$5 - \$3.95 + \$6 = \7.05
- I end up with: $\$11.85 - \$6 = \$5.85$
- Both better off, sender gains \$2.05, I gain \$5.85

If I return \$5:

- Sender ends up with: $\$5 - \$3.95 + \$5 = \6.05
- I end up with: $\$11.85 - \$5 = \$6.85$
- Sender gains $\$1.05$, I gain $\$6.85$

If I return \$7:

- Sender ends up with: $\$5 - \$3.95 + \$7 = \8.05
- I end up with: $\$11.85 - \$7 = \$4.85$
- Sender gains $\$3.05$, I gain $\$4.85$

A common approach in trust games is to split the gains somewhat fairly.

The sender's trust created \$7.90 in additional value beyond what they sent. An equal split of the total pool would give each of us \$6.45, meaning I'd return \$5.40. However, since the sender took the initial risk, returning \$6 seems more appropriate-it rewards their trust while still leaving me with a meaningful gain....

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 844,
  "reasoning_tokens": 0
}
```

message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.85000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience from Round 1:

- The previous sender showed strong trust by sending 79% of their endowment
- I reciprocated by returning about 50% of what I received
- This created mutual benefit for both of us

I want to maintain this cooperative dynamic. Since trust was shown to me and rewarded, I should demonstrate similar trust as the sender.

I'll send \$4.00 (80% of my endowment), which mirrors the high level of trust shown to me in Round 1. This signals my willingness to cooperate and hopefully encourages reciprocity.

```
{"send": 4.00}
```

reasoning:

Now I'm the SENDER in round 2.

Looking at my previous experience:

- In Round 1, I was the receiver
- The sender sent me \$3.95 (79% of their endowment)
- I received \$11.85 and returned \$6.00
- This worked out well for both of us - they gained \$2.05 and I gained \$5.85

Now I'm in the sender position. I should consider:

1. The previous sender showed high trust (79%)
2. I reciprocated fairly by returning about 50% of what I received
3. This created mutual benefit

Since the previous sender trusted me with 79% of their endowment and it worked out well, I should also show substantial trust. This could encourage cooperation and build a good reciprocal relationship.

Given that there may be communication noise, I should send a clear, substantial amount that demonstrates trust.

I'll send around \$4.00 (80% of my endowment), which is similar to the trust level shown to me in the previous round. This shows I'm willing to cooperate and trust the other player.

If they return a similar proportion as I did (roughly 50% of what they receive), I would get back around \$6, ending up with \$7 total, which would be a gain. Even if they return less, showing trust may encourage better long-term cooperation across rounds.

usage:

```
{
  "input_tokens": 634,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.850000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds.

I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

```
{"return": 0}
```

reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 910,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.270000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

{"return": 0}

reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85
- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
- So I must have received $5.85 - 1 = 0.85$? That seems low. Let me think...
- I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
- If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2
- In R2: $5 - 4 + \text{return} = \text{gain}$
- So $1 + \text{return} = \text{gain}$
- And $5.85 + \text{gain} = 6.85$
- So $\text{gain} = 1$
- So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

usage:

```
{
  "input_tokens": 1196,
```

```
"output_tokens": 746,  
"reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.270000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern across rounds:

- Round 2: I sent \$4 as sender, received very little back (lost money)
- Round 3: As receiver, got nothing sent to me
- Round 4: I sent \$2.50 as sender, received very little back (lost money again)
- Round 5: As receiver, got nothing sent to me again

The cooperation that existed in Round 1 has completely disappeared. Every time I've sent money as the sender, I've lost money. Every recent round as receiver, I've received nothing.

While the communication noise mentioned in the instructions is a consideration, I have to make decisions based on what I observe. The pattern is clear and consistent: sending money results in losses.

I need to protect my remaining earnings at this point.

```
{"send": 0}
```

reasoning:

Let me review what has happened:

Round 1: I was receiver, got \$11.85, returned \$6. The sender had sent \$3.95.

Round 2: I was sender, sent \$4, but the next round my earnings were only \$6.85, which means I got \$1 back (I had \$5 - \$4 + return = earnings - previous earnings = \$6.85 - \$5.85 = \$1). So I sent \$4, got \$1 back.

Round 3: I was receiver, got \$0. So the sender sent \$0.

Round 4: I was sender, sent \$2.50. My earnings went from \$6.85 to \$10.27, an increase of \$3.42. So I kept \$5 - \$2.50 + return = \$2.50 + return, and the increase was \$3.42, so return = \$3.42 - \$2.50 = \$0.92. Hmm, wait, let me recalculate. After round 3, I had \$6.85. In round 4 as sender, I start with \$5, send \$2.50, leaving me \$2.50. The return I got was such that my total earnings increased by \$10.27 - \$6.85 = \$3.42. So I earned \$2.50 (what I kept) + return, and this equals \$3.42, so return = \$0.92.

Round 5: I was receiver, got \$0.

So the pattern is:

- Round 1: High cooperation (someone sent me a lot, I returned fairly)
- Round 2: I sent \$4, got only \$1 back (poor return)
- Round 3: I got nothing sent to me
- Round 4: I sent \$2.50, got \$0.92 back (very poor return)
- Round 5: Again received nothing

The other player has consistently failed to reciprocate. My attempts to send money have resulted in substantial losses-\$3 net in round 2 and \$1.58 in round 4. The mention of communication noise raises a possibility: maybe what I intended to send didn't reach them, or their returns didn't reach me. But regardless, I'm facing round 6 as sender with a clear pattern of minimal cooperation. Sending \$0 protects my current earnings but forfeits any potential mutual gain.

There's another angle worth considering-if I'm paired with different people each round, then this person's actions shouldn't be judged by previous rounds' outcomes. The instructions suggest varying roles across rounds, which implies varying partners too. That would make each interaction somewhat fresh rather than a continuation of the same broken dynamic.

Looking at the pattern either way though: whether it's one person or five, I'm seeing consistent poor returns when I cooperate. Four of five rounds show weak reciprocation. The risk-reward calculation keeps pointing toward minimal sending. A \$1 bet might be the compromise-it caps my downside while leaving room for a positive surprise....

```
usage:
{
  "input_tokens": 1304,
  "output_tokens": 1383,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.270000000000001.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Assistant output:

The sender sent a very small amount (\$0.05, or 1% of their endowment), giving me \$0.15.

After several rounds of zero cooperation, this tiny gesture might be an attempt to test the waters and restart some level of trust. While it's minimal, it's the first non-zero amount in recent rounds.

I'll return a portion that rewards this small gesture and signals I'm open to rebuilding cooperation, without giving up everything from such a minimal initial investment.

```
{"return": 0.10}
```

Reasoning:

The sender sent \$0.05, which is a very small amount (only 1% of their endowment). I received \$0.15.

Looking at the pattern:

- After several rounds of zero cooperation, the sender is making a tiny gesture
- This could be a signal to try to restart cooperation
- Or it could just be a minimal effort

The sender currently has: \$5 - \$0.05 = \$4.95

If I return nothing, they end with \$4.95 (lost \$0.05)

If I return \$0.15, they end with \$5.10 (gained \$0.10)

If I return some middle amount like \$0.10, they end with \$5.05 (break even or small gain)

Given:

1. This is a very small gesture (only 1% sent)
2. But it's the first non-zero amount in recent rounds, possibly testing the waters
3. I want to signal willingness to cooperate if they show trust

I think I should return a proportional amount. Since they sent 1% of their endowment, which showed minimal trust, I'll return a modest amount that still rewards their small gesture.

Returning \$0.10 would give them:

- Sender: \$4.95 + \$0.10 = \$5.05 (small gain)
- Me: \$0.15 - \$0.10 = \$0.05

Actually, let me be a bit more generous to encourage cooperation. Returning \$0.12 would:

- Sender: \$4.95 + \$0.12 = \$5.07 (small gain)
- Me: \$0.15 - \$0.12 = \$0.03

This shows I'm willing to reciprocate even small gestures, potentially restarting cooperation.

```
Usage: {
  "input_tokens": 1565,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

```

dyad_1: Agent_2 sent $1.5; received $4.5; Agent_1 returned $0.3; payoffs Agent_2=$3.8,
Agent_1=$4.2; noise/visible: sent visible=$0.15, received visible=$0.45, returned visible=$0
Action validation: {"returned": {"amount": 0.3, "clamped": false, "max": 4.5, "min": 0, "raw":
0.3}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}
Balances after round: {
  "Agent_1": 39.800000000000004,
  "Agent_2": 36.199999999999996
}
Visible balances after round: {
  "Agent_1": 17.43,
  "Agent_2": 18.82
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:13:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You are the RECEIVER this round. Your total earnings across all rounds are \$6.850000000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds. I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

{"return": 0}

reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds where there was high trust and cooperation.

Since I received \$0, I have nothing to return - the maximum I can return is \$0 (as stated in the prompt).

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

Either way, I cannot return anything because I received nothing.

My total earnings remain at \$6.85 from previous rounds.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.850000000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: As receiver, someone trusted me with 79%, I returned fairly
- Round 2: I trusted the receiver with 80%, but received nothing back (lost \$4)
- Round 3: As receiver, I got nothing sent to me

The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I want to signal willingness to cooperate while being cautious given recent outcomes. Sending nothing perpetuates the breakdown, but sending too much risks another significant loss.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

{"send": 2.50}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00

Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...

Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

My total earnings are \$6.85.

Let me trace through more carefully:

- Round 1 ended: I had \$5.85 (started with \$0, received \$11.85, returned \$6.00)
- Round 2: I sent \$4.00 as sender, so I had \$1.00 left plus whatever was returned to me
 - If my total is now \$6.85, that means in Round 2, I ended with \$6.85 - \$5.85 = \$1.00
 - I started Round 2 with \$5, sent \$4.00, kept \$1.00
 - I received back \$0 (since \$1.00 + \$0 = \$1.00)
 - So the receiver in Round 2 got \$12.00 (my \$4 × 3) and returned \$0!
- Round 3: I was receiver, got \$0, so my earnings stayed at \$6.85

So what happened:

- Round 1: Sender trusted me (79%), I reciprocated fairly (returned ~50%)
- Round 2: I trusted the receiver (80%), they betrayed me completely (returned 0%)
- Round 3: New sender didn't trust me at all (sent 0%)

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 910,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.270000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

{"return": 0}

reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85
- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
 - So I must have received 5.85 - 1 = 0.85? That seems low. Let me think...
 - I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
 - If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2

- In R2: $5 - 4 + \text{return} = \text{gain}$
- So $1 + \text{return} = \text{gain}$
- And $5.85 + \text{gain} = 6.85$
- So $\text{gain} = 1$
- So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.2700000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern across rounds:

- Round 2: I sent \$4 as sender, received very little back (lost money)
- Round 3: As receiver, got nothing sent to me
- Round 4: I sent \$2.50 as sender, received very little back (lost money again)
- Round 5: As receiver, got nothing sent to me again

The cooperation that existed in Round 1 has completely disappeared. Every time I've sent money as the sender, I've lost money. Every recent round as receiver, I've received nothing.

While the communication noise mentioned in the instructions is a consideration, I have to make decisions based on what I observe. The pattern is clear and consistent: sending money results in losses.

I need to protect my remaining earnings at this point.

```
{"send": 0}
```

reasoning:

Let me review what has happened:

Round 1: I was receiver, got \$11.85, returned \$6. The sender had sent \$3.95.

Round 2: I was sender, sent \$4, but the next round my earnings were only \$6.85, which means I got \$1 back ($5 - 4 + \text{return} = \text{earnings} - \text{previous earnings} = 6.85 - 5.85 = 1$). So I sent \$4, got \$1 back.

Round 3: I was receiver, got \$0. So the sender sent \$0.

Round 4: I was sender, sent \$2.50. My earnings went from \$6.85 to \$10.27, an increase of \$3.42. So I kept $5 - 2.50 + \text{return} = 2.50 + \text{return}$, and the increase was \$3.42, so $\text{return} = 3.42 - 2.50 = 0.92$. Hmm, wait, let me recalculate. After round 3, I had \$6.85.

In round 4 as sender, I start with \$5, send \$2.50, leaving me \$2.50. The return I got was such that my total earnings increased by $10.27 - 6.85 = 3.42$. So I earned \$2.50 (what I kept) + return, and this equals \$3.42, so $\text{return} = 0.92$.

Round 5: I was receiver, got \$0.

So the pattern is:

- Round 1: High cooperation (someone sent me a lot, I returned fairly)
- Round 2: I sent \$4, got only \$1 back (poor return)
- Round 3: I got nothing sent to me
- Round 4: I sent \$2.50, got \$0.92 back (very poor return)
- Round 5: Again received nothing

The other player has consistently failed to reciprocate. My attempts to send money have resulted in substantial losses-\$3 net in round 2 and \$1.58 in round 4. The mention of comm

unication noise raises a possibility: maybe what I intended to send didn't reach them, or their returns didn't reach me. But regardless, I'm facing round 6 as sender with a clear pattern of minimal cooperation. Sending \$0 protects my current earnings but forfeits any potential mutual gain.

There's another angle worth considering-if I'm paired with different people each round, then this person's actions shouldn't be judged by previous rounds' outcomes. The instructions suggest varying roles across rounds, which implies varying partners too. That would make each interaction somewhat fresh rather than a continuation of the same broken dynamic.

Looking at the pattern either way though: whether it's one person or five, I'm seeing consistent poor returns when I cooperate. Four of five rounds show weak reciprocation. The risk-reward calculation keeps pointing toward minimal sending. A \$1 bet might be the compromise-it caps my downside while leaving room for a positive surprise....

usage:

```
{
  "input_tokens": 1304,
  "output_tokens": 1383,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.270000000000001.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 13 role=assistant

content:

The sender sent a very small amount (\$0.05, or 1% of their endowment), giving me \$0.15.

After several rounds of zero cooperation, this tiny gesture might be an attempt to test the waters and restart some level of trust. While it's minimal, it's the first non-zero amount in recent rounds.

I'll return a portion that rewards this small gesture and signals I'm open to rebuilding cooperation, without giving up everything from such a minimal initial investment.

```
{"return": 0.10}
```

reasoning:

The sender sent \$0.05, which is a very small amount (only 1% of their endowment). I received \$0.15.

Looking at the pattern:

- After several rounds of zero cooperation, the sender is making a tiny gesture
- This could be a signal to try to restart cooperation
- Or it could just be a minimal effort

The sender currently has: \$5 - \$0.05 = \$4.95

If I return nothing, they end with \$4.95 (lost \$0.05)

If I return \$0.15, they end with \$5.10 (gained \$0.10)

If I return some middle amount like \$0.10, they end with \$5.05 (break even or small gain)

Given:

1. This is a very small gesture (only 1% sent)
2. But it's the first non-zero amount in recent rounds, possibly testing the waters
3. I want to signal willingness to cooperate if they show trust

I think I should return a proportional amount. Since they sent 1% of their endowment, which showed minimal trust, I'll return a modest amount that still rewards their small gesture.

Returning \$0.10 would give them:

- Sender: \$4.95 + \$0.10 = \$5.05 (small gain)
- Me: \$0.15 - \$0.10 = \$0.05

Actually, let me be a bit more generous to encourage cooperation. Returning \$0.12 would:

- Sender: $\$4.95 + \$0.12 = \$5.07$ (small gain)
- Me: $\$0.15 - \$0.12 = \$0.03$

This shows I'm willing to reciprocate even small gestures, potentially restarting cooperation.

usage:

```
{
  "input_tokens": 1565,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.320000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at Round 7:

- The sender made a small gesture by sending \$0.05
- I responded positively by returning \$0.10 (more than half of what I received)
- This was our first positive interaction in several rounds

This small exchange might be a signal to rebuild cooperation after the breakdown in rounds 2-6. My generous return in Round 7 showed I'm willing to cooperate.

However, I need to be cautious. The last two times I sent substantial amounts as sender (Rounds 2 and 4), I lost money.

I'll send a moderate amount that signals willingness to cooperate while managing risk. \$1.50 (30% of endowment) is meaningful but measured - enough to test if we can rebuild trust without risking too much.

```
{"send": 1.50}
```

Reasoning:

Let me review what happened in Round 7:

- As receiver, I got \$0.15 (sender sent \$0.05)
- I returned \$0.10 to the sender

This was the first positive interaction in several rounds. The sender sent a tiny amount as a test, and I responded generously by returning most of what I received.

Now I'm the sender again in Round 8. The question is whether this small olive branch leads to rebuilding trust.

Given:

- The cooperation completely broke down in rounds 2-6
- Round 7 showed a tiny test from the sender
- I responded positively to that test
- This might be an opportunity to rebuild cooperation

However, I need to be careful. The last two times I sent substantial amounts (Round 2: \$4, Round 4: \$2.50), I lost money both times.

I think I should send a moderate amount - more than the tiny \$0.05 test, but not the large amounts I sent before. This would:

1. Signal I'm willing to cooperate after the positive Round 7 interaction
2. Not risk too much given the history
3. Test if the receiver is actually ready to cooperate

I'll send \$1.50 (30% of endowment). This is a meaningful amount that shows trust without being too risky.

Usage: {

```
"input_tokens": 1480,
"output_tokens": 491,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:13:47
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at this situation:

****Round 2 Status:****

- I'm now the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, I have no funds to return. The only valid response is:

```
'''json
{'return': 0}
'''
```

****Note:**** I see from my total earnings (\$7.02) that in Round 1, the receiver was quite generous with their return. Unfortunately, this round the sender chose not to send anything, so there's nothing I can return even if I wanted to reciprocate that previous generosity.

reasoning:

Let me understand this situation:

Round 1: I was the SENDER and sent \$3

- I started with \$5
- I sent \$3 (so the receiver got \$9)
- My total earnings are now \$7.02

Let me work backwards:

- I started with \$5
- I sent \$3, so I had \$2 left
- Total earnings are \$7.02
- So I must have received: $\$7.02 - \$2 = \$5.02$ back from the receiver

So in Round 1, the receiver returned \$5.02 out of the \$9 they received. That's quite generous!

Round 2: I am now the RECEIVER

- The sender sent me \$0
- So I received \$0 (since $\$0 \times 3 = \0)
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I didn't receive anything.

Wait, let me reconsider the earnings calculation. It says "Your total earnings across all rounds are \$7.02"

Actually, I think this means my cumulative total across all rounds so far is \$7.02. Let me recalculate:

Round 1 (as SENDER):

- Started with \$5
- Sent \$3 (kept \$2)
- Receiver got \$9 ($3 \times \3)
- Receiver returned some amount X
- My round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total earnings are \$7.02 after round 1, then:

$$\$2 + X = \$7.02$$

$$X = \$5.02$$

So the receiver returned \$5.02 to me in round 1. That's very generous - they kept \$3.98 out of \$9 and returned \$5.02.

Now in Round 2:

...

usage:

```
{
  "input_tokens": 593,
  "output_tokens": 656,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 3

  You are the SENDER this round. Your total earnings across all rounds are $7.02.
  How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
  Looking at the situation:

  **Previous rounds:**
  - Round 1: I sent $3, and based on my earnings ($7.02), the receiver generously returned a
    bout $5, keeping less than half of what they received
  - Round 2: I was the receiver, got $0, so returned $0

  **Round 3 Analysis:**
  - The Round 1 receiver showed strong reciprocity and trustworthiness
  - Round 2's sender (potentially a different person) sent nothing
  - I'm now paired with possibly another new partner

  **My decision:**
  Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to
  trust and build potential long-term cooperation. Sending a moderate amount:
  - Tests if this receiver is trustworthy like Round 1's
  - Creates opportunity for mutual gain through the 3x multiplier
  - Maintains some security by not sending everything

  ```json
 {'send': 3}
  ```

reasoning:
  Let me think about this situation:

  Round 1: I was the SENDER and sent $3
  - I should have kept $5 - $3 = $2
  - The receiver got $3 × 3 = $9
  - My total earnings show $7.02

  If I kept $2 from Round 1, and now have total earnings of $7.02, that means I received bac
  k $5.02 in Round 1.
  That's a very generous return! The receiver returned $5.02 out of the $9 they received.

  Round 2: I was the RECEIVER
  - The sender sent me $0
  - I received $0 × 3 = $0
  - I returned $0
  - So Round 2 added $0 to my earnings

  My total is still $7.02, which confirms this.

  Now in Round 3, I'm the SENDER again.

  Analysis:
  - In Round 1, when I sent $3, the receiver was very generous and returned $5.02 out of $9
    (keeping only $3.98)
  - In Round 2, when I was the receiver, the sender gave me nothing
  - This might be a different person, or the same person being less generous

  Given that:
  1. The first receiver was very trustworthy and generous
  2. This is a multi-round game where building trust matters
  3. The sender in Round 2 might be testing me or might be the ungenerous player
  4. I should probably try to cooperate to maximize joint welfare

  I think I should send a decent amount again. The fact that the Round 1 receiver was so gen
  erous suggests there are trustworthy players in this experiment. I should try to identif
  y and cooperate with them.

```

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

```
**Round 4 Status:**
- I'm the RECEIVER
- The sender sent me $0
- After the 3x multiplier:  $\$0 \times 3 = \$0$ 
- I received nothing
```

Since I received \$0, there are no funds available to return. The only possible response is :

```
'''json
{'return': 0}
'''
```

****Reflection:**** This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.

After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

```
**Round 5 Analysis:**
```

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return

- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate
- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: \$2 + \$5.02 = \$7.02
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: \$2 + \$0.81 = \$2.81
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

...

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.1.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 11 role=assistant

content:

****Round 6 Status:****

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: \$0.15 received

****My decision:****

The sender sent a very small amount, but they still showed some trust by sending something rather than nothing. Given that I've received \$0 in most rounds as receiver, even a small gesture of cooperation is notable.

To signal that I'm trustworthy and encourage future cooperation, I'll return a fair portion that shares the gains from the multiplier:

- If I return \$0.10, the sender nets \$0.05 profit overall
- I keep \$0.05
- This demonstrates I'm willing to cooperate even with small amounts

```
'''json
{'return': 0.10}
'''
```

This way, both of us benefit from the interaction, and I establish myself as a reliable partner for potential future rounds.

reasoning:

Round 6 Analysis:

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: $\$0.05 \times 3 = \0.15
- My total earnings are \$13.1

This is interesting - the sender sent a very small amount (\$0.05). This could be:

1. A test to see if I'm trustworthy with small amounts
2. The sender being very cautious after bad experiences
3. Communication noise (though \$0.05 seems deliberate)

Given that:

- The amount is tiny (\$0.15 received)
- The sender showed some trust, even if minimal
- I've been getting mostly \$0 as a receiver, so any trust gesture is notable
- I should establish myself as trustworthy if I want future cooperation

I think I should return a fair amount or even slightly more than "fair" to signal trustworthiness. A typical fair split might be 50-50, so I'd return \$0.075 and keep \$0.075.

But given the pattern of non-cooperation, maybe I should return slightly more to build trust - perhaps return \$0.10 and keep \$0.05? This would show I'm cooperative even with small amounts.

Actually, let me think about this differently. The sender:

- Kept \$4.95
- Gave me \$0.05 which became \$0.15
- If I return \$0.10, they end up with $\$4.95 + \$0.10 = \$5.05$ (gained \$0.05)
- I end up with \$0.05

This seems reasonable and fair - I'm sharing the gains from the multiplier. Let me go with returning \$0.10.

usage:

```
{
  "input_tokens": 1556,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.25.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

****Round 7 Analysis:****

Looking at my recent experience:

- Round 5 (Sender): Sent \$2 ? received minimal return
- Round 6 (Receiver): Got only \$0.05, but I returned \$0.10 generously to encourage cooperation

****Current situation:****

My earnings have been growing, but cooperation as sender has been hit-or-miss. However, in Round 6 I demonstrated that I'm a trustworthy receiver by returning more than I kept even from a tiny amount.

****My decision:****

I'll continue testing for cooperative partners. The 3x multiplier creates real potential for mutual benefit. I'll send a moderate amount:

I'll send ****\$2****:

- Shows willingness to build trust
- Creates \$6 for receiver (good potential for both of us)
- Keeps \$3 as security
- Balances cooperation with risk management

```
'''json
{'send': 2}
'''
```

If more receivers prove trustworthy, I might increase; if cooperation continues to fail, I'll be more conservative.

reasoning:

Let me track my earnings across rounds:

Round 1 (Sender): Sent \$3, kept \$2, got generous return ? ended around \$7.02
Round 2 (Receiver): Got \$0, returned \$0 ? still \$7.02
Round 3 (Sender): Sent \$3, kept \$2, got minimal return (~\$0.81) ? ended around \$9.83
Round 4 (Receiver): Got \$0, returned \$0 ? still \$9.83
Round 5 (Sender): Sent \$2, kept \$3, got return of ~\$0.27 ? ended around \$13.10
Round 6 (Receiver): Got \$0.15, returned \$0.10, kept \$0.05 ? ended around \$13.15
Now Round 7: Starting at \$13.25 (gained \$0.10 from Round 5 return it seems)

Wait, let me recalculate:

- Round 5 total was \$9.83
- Round 6 started at \$13.1, which is \$3.27 higher
- This suggests in Round 5, I sent \$2, kept \$3, and got back around \$3.27? That's a very good return!
- Round 6: I was receiver, got \$0.15, returned \$0.10, kept \$0.05 ? $\$13.1 + \$0.05 = \$13.15$
- Current earnings: \$13.25, so I gained \$0.10 from being sender in some recent round

Actually, looking at the pattern:

- Round 4 ended at \$9.83
- Round 5 I was sender and my earnings should have been: $\$9.83 - \text{what I sent} + \text{what was returned} + \5 new
- Round 6 started at \$13.1

If I sent \$2 in Round 5:

- $\$9.83 - \$2 + \text{return} + \$5 = \13.1
- $\text{return} = \$13.1 - \$9.83 - \$5 + \$2 = \$0.27$

Then Round 6: $\$13.1 + \$0.05 \text{ (kept from } \$0.15) = \$13.15$

Round 7 start: \$13.25

There's a \$0.10 gap I haven't fully accounted for-possibly a small return from an earlier round. The clearer pattern emerging is that when I send money, returns vary dramatically: the first round was generous, the third minimal, and the fifth modest. As a receiver, I've gotten almost nothing back....

usage:

```
{
  "input_tokens": 1803,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.28.

This round, The sender sent you \$0.15, so you received \$0.4499999999999996. How much do you return to The sender? (0-0.4499999999999996)

Assistant output:

****Round 8 Status:****

- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****My decision:****

Similar to Round 6, the sender sent a small amount but showed trust. To continue building a reputation as a fair and trustworthy partner, I'll share the gains from the multiplier equitably.

The multiplier created \$0.30 in extra value (\$0.15 * \$0.45). A fair split would give us each \$0.15 in gains:

- If I return \$0.30, the sender nets +\$0.15 overall
- I keep \$0.15
- Both of us benefit equally

This demonstrates consistent trustworthiness and encourages future cooperation.

```
'''json
{'return': 0.30}
'''
```

Reasoning:

****Round 8 Status:****

- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****Analysis:****

The sender sent a very small amount again (\$0.15), similar to Round 6. This could be:

1. Testing me to see if I'm trustworthy
2. Being cautious due to past experiences
3. Communication noise affecting what they see

In Round 6, I demonstrated trustworthiness by returning \$0.10 out of \$0.15 received. I should continue that pattern to:

- Establish myself as a reliable partner
- Encourage future cooperation
- Share the gains from the 3x multiplier fairly

****Return calculation:****

- Received: \$0.45
- If I return \$0.30:
 - Sender profit: $-\$0.15 + \$0.30 = \$0.15$ net gain
 - I keep: \$0.15
 - We both gain equally from the interaction

This seems fair and demonstrates I'm willing to cooperate even with small amounts. It also splits the multiplier benefit evenly.

Actually, let me recalculate more carefully:

- Sender sent: \$0.15
- I received: \$0.45
- Total value created by multiplier: $\$0.15 * \0.45 , so \$0.30 gain
- Fair split: we each get \$0.15 of the gain
- So sender should end with: $\$5 - \$0.15 + \text{return}$
- For sender to gain \$0.15: $\text{return} = \$0.30$
- I would keep: $\$0.45 - \$0.30 = \$0.15$

...

```
Usage: {
  "input_tokens": 1919,
  "output_tokens": 614,
  "reasoning_tokens": 0
}
```

}

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.49

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 42.800000000000004,

"Agent_2": 42.199999999999996

}

Visible balances after round: {

"Agent_1": 20.92,

"Agent_2": 18.82

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:14:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.02.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the situation:

****Previous rounds:****

- Round 1: I sent \$3, and based on my earnings (\$7.02), the receiver generously returned about \$5, keeping less than half of what they received
- Round 2: I was the receiver, got \$0, so returned \$0

****Round 3 Analysis:****

- The Round 1 receiver showed strong reciprocity and trustworthiness
- Round 2's sender (potentially a different person) sent nothing
- I'm now paired with possibly another new partner

****My decision:****

Given that cooperation proved beneficial in Round 1, I'll continue showing willingness to trust and build potential long-term cooperation. Sending a moderate amount:

- Tests if this receiver is trustworthy like Round 1's
- Creates opportunity for mutual gain through the 3x multiplier
- Maintains some security by not sending everything

```
'''json
{'send': 3}
'''
```

reasoning:

Let me think about this situation:

Round 1: I was the SENDER and sent \$3

- I should have kept \$5 - \$3 = \$2
- The receiver got $\$3 \times 3 = \9
- My total earnings show \$7.02

If I kept \$2 from Round 1, and now have total earnings of \$7.02, that means I received back \$5.02 in Round 1.

That's a very generous return! The receiver returned \$5.02 out of the \$9 they received.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received $\$0 \times 3 = \0
- I returned \$0
- So Round 2 added \$0 to my earnings

My total is still \$7.02, which confirms this.

Now in Round 3, I'm the SENDER again.

Analysis:

- In Round 1, when I sent \$3, the receiver was very generous and returned \$5.02 out of \$9 (keeping only \$3.98)
- In Round 2, when I was the receiver, the sender gave me nothing
- This might be a different person, or the same person being less generous

Given that:

1. The first receiver was very trustworthy and generous
2. This is a multi-round game where building trust matters
3. The sender in Round 2 might be testing me or might be the ungenerous player
4. I should probably try to cooperate to maximize joint welfare

I think I should send a decent amount again. The fact that the Round 1 receiver was so generous suggests there are trustworthy players in this experiment. I should try to identify and cooperate with them.

I'll send \$3 again - it's a fair middle ground that signals cooperation while not being reckless.

usage:

```
{
  "input_tokens": 788,
  "output_tokens": 636,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

```
**Round 4 Status:**
- I'm the RECEIVER
- The sender sent me $0
- After the 3x multiplier:  $\$0 \times 3 = \$0$ 
- I received nothing
```

Since I received \$0, there are no funds available to return. The only possible response is :

```
'''json
{'return': 0}
'''
```

****Reflection:**** This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.

After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
```

```
"input_tokens": 1057,
"output_tokens": 394,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

****Round 5 Analysis:****

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return
- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate
- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: \$2 + \$5.02 = \$7.02
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: \$2 + \$0.81 = \$2.81
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

```
...
usage:
{
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.1.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 9 role=assistant

content:

****Round 6 Status:****

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: \$0.15 received

****My decision:****

The sender sent a very small amount, but they still showed some trust by sending something rather than nothing. Given that I've received \$0 in most rounds as receiver, even a small gesture of cooperation is notable.

To signal that I'm trustworthy and encourage future cooperation, I'll return a fair portion that shares the gains from the multiplier:

- If I return \$0.10, the sender nets \$0.05 profit overall
- I keep \$0.05
- This demonstrates I'm willing to cooperate even with small amounts

```
'''json
{'return': 0.10}
'''
```

This way, both of us benefit from the interaction, and I establish myself as a reliable partner for potential future rounds.

reasoning:

Round 6 Analysis:

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: $\$0.05 \times 3 = \0.15
- My total earnings are \$13.1

This is interesting - the sender sent a very small amount (\$0.05). This could be:

1. A test to see if I'm trustworthy with small amounts
2. The sender being very cautious after bad experiences
3. Communication noise (though \$0.05 seems deliberate)

Given that:

- The amount is tiny (\$0.15 received)
- The sender showed some trust, even if minimal
- I've been getting mostly \$0 as a receiver, so any trust gesture is notable
- I should establish myself as trustworthy if I want future cooperation

I think I should return a fair amount or even slightly more than "fair" to signal trustworthiness. A typical fair split might be 50-50, so I'd return \$0.075 and keep \$0.075.

But given the pattern of non-cooperation, maybe I should return slightly more to build trust - perhaps return \$0.10 and keep \$0.05? This would show I'm cooperative even with small amounts.

Actually, let me think about this differently. The sender:

- Kept \$4.95
- Gave me \$0.05 which became \$0.15
- If I return \$0.10, they end up with $\$4.95 + \$0.10 = \$5.05$ (gained \$0.05)
- I end up with \$0.05

This seems reasonable and fair - I'm sharing the gains from the multiplier. Let me go with returning \$0.10.

usage:

```
{
  "input_tokens": 1556,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.25.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

****Round 7 Analysis:****

Looking at my recent experience:

- Round 5 (Sender): Sent \$2 ? received minimal return
- Round 6 (Receiver): Got only \$0.05, but I returned \$0.10 generously to encourage cooperation

****Current situation:****

My earnings have been growing, but cooperation as sender has been hit-or-miss. However, in Round 6 I demonstrated that I'm a trustworthy receiver by returning more than I kept even from a tiny amount.

****My decision:****

I'll continue testing for cooperative partners. The 3x multiplier creates real potential for mutual benefit. I'll send a moderate amount:

I'll send ****\$2****:

- Shows willingness to build trust
- Creates \$6 for receiver (good potential for both of us)
- Keeps \$3 as security
- Balances cooperation with risk management

```
'''json
{'send': 2}
'''
```

If more receivers prove trustworthy, I might increase; if cooperation continues to fail, I'll be more conservative.

reasoning:

Let me track my earnings across rounds:

Round 1 (Sender): Sent \$3, kept \$2, got generous return ? ended around \$7.02
Round 2 (Receiver): Got \$0, returned \$0 ? still \$7.02
Round 3 (Sender): Sent \$3, kept \$2, got minimal return (~\$0.81) ? ended around \$9.83
Round 4 (Receiver): Got \$0, returned \$0 ? still \$9.83
Round 5 (Sender): Sent \$2, kept \$3, got return of ~\$0.27 ? ended around \$13.10
Round 6 (Receiver): Got \$0.15, returned \$0.10, kept \$0.05 ? ended around \$13.15
Now Round 7: Starting at \$13.25 (gained \$0.10 from Round 5 return it seems)

Wait, let me recalculate:

- Round 5 total was \$9.83
- Round 6 started at \$13.1, which is \$3.27 higher
- This suggests in Round 5, I sent \$2, kept \$3, and got back around \$3.27? That's a very good return!
- Round 6: I was receiver, got \$0.15, returned \$0.10, kept \$0.05 ? $\$13.1 + \$0.05 = \$13.15$
- Current earnings: \$13.25, so I gained \$0.10 from being sender in some recent round

Actually, looking at the pattern:

- Round 4 ended at \$9.83
- Round 5 I was sender and my earnings should have been: $\$9.83 - \text{what I sent} + \text{what was returned} + \5 new
- Round 6 started at \$13.1

If I sent \$2 in Round 5:

- \$9.83 - \$2 + return + \$5 = \$13.1
- return = \$13.1 - \$9.83 - \$5 + \$2 = \$0.27

Then Round 6: \$13.1 + \$0.05 (kept from \$0.15) = \$13.15
Round 7 start: \$13.25

There's a \$0.10 gap I haven't fully accounted for-possibly a small return from an earlier round. The clearer pattern emerging is that when I send money, returns vary dramatically : the first round was generous, the third minimal, and the fifth modest. As a receiver, I've gotten almost nothing back....

usage:

```
{
  "input_tokens": 1803,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.28. This round, The sender sent you \$0.15, so you received \$0.44999999999999996. How much do you return to The sender? (0-0.44999999999999996)

Message 13 role=assistant

content:

****Round 8 Status:****
- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****My decision:****

Similar to Round 6, the sender sent a small amount but showed trust. To continue building a reputation as a fair and trustworthy partner, I'll share the gains from the multiplier equitably.

The multiplier created \$0.30 in extra value (\$0.15 * 3 = \$0.45). A fair split would give us each \$0.15 in gains:

- If I return \$0.30, the sender nets +\$0.15 overall
- I keep \$0.15
- Both of us benefit equally

This demonstrates consistent trustworthiness and encourages future cooperation.

```
'''json
{'return': 0.30}
'''
```

reasoning:

****Round 8 Status:****
- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****Analysis:****

The sender sent a very small amount again (\$0.15), similar to Round 6. This could be:

1. Testing me to see if I'm trustworthy
2. Being cautious due to past experiences
3. Communication noise affecting what they see

In Round 6, I demonstrated trustworthiness by returning \$0.10 out of \$0.15 received. I should continue that pattern to:

- Establish myself as a reliable partner
- Encourage future cooperation
- Share the gains from the 3x multiplier fairly

****Return calculation:****

- Received: \$0.45
- If I return \$0.30:
 - Sender profit: -\$0.15 + \$0.30 = \$0.15 net gain
 - I keep: \$0.15
 - We both gain equally from the interaction

This seems fair and demonstrates I'm willing to cooperate even with small amounts. It also splits the multiplier benefit evenly.

Actually, let me recalculate more carefully:

- Sender sent: \$0.15
- I received: \$0.45
- Total value created by multiplier: $\$0.15 \times 3 = \0.45 , so \$0.30 gain
- Fair split: we each get \$0.15 of the gain
- So sender should end with: $\$5 - \$0.15 + \text{return}$
- For sender to gain \$0.15: $\text{return} = \$0.30$
- I would keep: $\$0.45 - \$0.30 = \$0.15$

...

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 614,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.43.

How much do you send to the receiver this round? (0-5)

Assistant output:

****Round 9 Analysis:****

Looking at my track record:

- As RECEIVER (Rounds 6 & 8): I've been consistently fair, returning more than I kept even from tiny amounts
- As SENDER (Rounds 7 & previous): Mixed results with cooperation

****Current earnings:**** \$17.43

****My decision:****

Despite inconsistent returns when I'm the sender, the 3x multiplier still creates genuine opportunity for mutual benefit. I've proven I'm trustworthy as a receiver - hopefully I'll encounter similarly cooperative partners when I'm the sender.

I'll send ****\$2****:

- Demonstrates continued willingness to cooperate
- Creates \$6 for the receiver
- Balances opportunity with risk management
- Consistent with my recent sending strategy

```
'''json
{'send': 2}
'''
```

The key to this game is finding and building relationships with cooperative partners, even if progress is gradual.

Reasoning:

Let me review the situation:

****My history:****

- Round 1 (Sender): Sent \$3 ? got very generous return
- Round 2 (Receiver): Got \$0 ? returned \$0
- Round 3 (Sender): Sent \$3 ? got minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0 ? returned \$0
- Round 5 (Sender): Sent \$2 ? got minimal return
- Round 6 (Receiver): Got \$0.05 (\$0.15 after 3x) ? returned \$0.10
- Round 7 (Sender): Sent \$2 ? earnings went from \$13.25 to \$17.28, so I gained about \$4.03 (that's \$5 - \$2 sent + return). So I got back about \$1.03 return from \$6 sent to them.
- Round 8 (Receiver): Got \$0.15 (\$0.45 after 3x) ? returned \$0.30

Now in Round 9: Earnings at \$17.43 (gained \$0.15 from Round 8)

****Pattern Analysis:****

- As a sender, results have been very mixed

- As a receiver, I've been consistently fair and trustworthy
- Recent rounds (6 and 8) involved very small amounts sent to me, but I returned fairly each time
- Round 7 sending \$2 didn't work out very well

****Considerations:****

- There may be communication noise mentioned in the instructions
- Different partners each round likely
- I've established myself as trustworthy when receiving
- As sender, I've had mixed success

I should continue trying to find cooperative partners. The game benefits everyone when there's trust. I'll send a moderate amount again.

I'll send \$2 again - it's a reasonable middle ground that shows willingness to cooperate while managing risk.

```
Usage: {
  "input_tokens": 1947,
  "output_tokens": 661,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:14:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.850000000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is quite different from the cooperative pattern we had established in earlier rounds.

I'm disappointed that the trust wasn't continued, but I can only work with what I receive.

{"return": 0}

reasoning:

The sender sent me \$0, so I received \$0. This is quite different from the previous rounds where there was high trust and cooperation.

Since I received \$0, I have nothing to return - the maximum I can return is \$0 (as stated in the prompt).

The sender chose not to send anything. This could be:

1. A strategic decision
2. A breakdown in trust
3. Perhaps they're testing me
4. Or there could be communication noise mentioned in the instructions

Either way, I cannot return anything because I received nothing.

My total earnings remain at \$6.85 from previous rounds.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 207,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.850000000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: As receiver, someone trusted me with 79%, I returned fairly
- Round 2: I trusted the receiver with 80%, but received nothing back (lost \$4)
- Round 3: As receiver, I got nothing sent to me

The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I want to signal willingness to cooperate while being cautious given recent outcomes. Sending nothing perpetuates the breakdown, but sending too much risks another significant loss.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

```
{"send": 2.50}
```

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00

Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...

Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

My total earnings are \$6.85.

Let me trace through more carefully:

- Round 1 ended: I had \$5.85 (started with \$0, received \$11.85, returned \$6.00)
- Round 2: I sent \$4.00 as sender, so I had \$1.00 left plus whatever was returned to me
 - If my total is now \$6.85, that means in Round 2, I ended with \$6.85 - \$5.85 = \$1.00
 - I started Round 2 with \$5, sent \$4.00, kept \$1.00
 - I received back \$0 (since \$1.00 + \$0 = \$1.00)
 - So the receiver in Round 2 got \$12.00 (my \$4 × 3) and returned \$0!
- Round 3: I was receiver, got \$0, so my earnings stayed at \$6.85

So what happened:

- Round 1: Sender trusted me (79%), I reciprocated fairly (returned ~50%)
- Round 2: I trusted the receiver (80%), they betrayed me completely (returned 0%)
- Round 3: New sender didn't trust me at all (sent 0%)

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

usage:

```
{  
  "input_tokens": 952,  
  "output_tokens": 910,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.270000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

```
{"return": 0}
```

reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85
- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
 - So I must have received $5.85 - 1 = 0.85$? That seems low. Let me think...
 - I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
 - If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2
 - In R2: $5 - 4 + \text{return} = \text{gain}$
 - So $1 + \text{return} = \text{gain}$
 - And $5.85 + \text{gain} = 6.85$
 - So $\text{gain} = 1$
 - So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.270000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

- Round 2: I sent \$4 as sender, received very little back (lost money)
- Round 3: As receiver, got nothing sent to me
- Round 4: I sent \$2.50 as sender, received very little back (lost money again)
- Round 5: As receiver, got nothing sent to me again

The cooperation that existed in Round 1 has completely disappeared. Every time I've sent money as the sender, I've lost money. Every recent round as receiver, I've received nothing.

While the communication noise mentioned in the instructions is a consideration, I have to make decisions based on what I observe. The pattern is clear and consistent: sending money results in losses.

I need to protect my remaining earnings at this point.

```
{"send": 0}
```

reasoning:

Let me review what has happened:

Round 1: I was receiver, got \$11.85, returned \$6. The sender had sent \$3.95.

Round 2: I was sender, sent \$4, but the next round my earnings were only \$6.85, which mean

s I got \$1 back (I had \$5 - \$4 + return = earnings - previous earnings = \$6.85 - \$5.85 = \$1). So I sent \$4, got \$1 back.
 Round 3: I was receiver, got \$0. So the sender sent \$0.
 Round 4: I was sender, sent \$2.50. My earnings went from \$6.85 to \$10.27, an increase of \$3.42. So I kept \$5 - \$2.50 + return = \$2.50 + return, and the increase was \$3.42, so return = \$3.42 - \$2.50 = \$0.92. Hmm, wait, let me recalculate. After round 3, I had \$6.85. In round 4 as sender, I start with \$5, send \$2.50, leaving me \$2.50. The return I got was such that my total earnings increased by \$10.27 - \$6.85 = \$3.42. So I earned \$2.50 (what I kept) + return, and this equals \$3.42, so return = \$0.92.
 Round 5: I was receiver, got \$0.

So the pattern is:

- Round 1: High cooperation (someone sent me a lot, I returned fairly)
- Round 2: I sent \$4, got only \$1 back (poor return)
- Round 3: I got nothing sent to me
- Round 4: I sent \$2.50, got \$0.92 back (very poor return)
- Round 5: Again received nothing

The other player has consistently failed to reciprocate. My attempts to send money have resulted in substantial losses-\$3 net in round 2 and \$1.58 in round 4. The mention of communication noise raises a possibility: maybe what I intended to send didn't reach them, or their returns didn't reach me. But regardless, I'm facing round 6 as sender with a clear pattern of minimal cooperation. Sending \$0 protects my current earnings but forfeits any potential mutual gain.

There's another angle worth considering-if I'm paired with different people each round, then this person's actions shouldn't be judged by previous rounds' outcomes. The instructions suggest varying roles across rounds, which implies varying partners too. That would make each interaction somewhat fresh rather than a continuation of the same broken dynamic.

Looking at the pattern either way though: whether it's one person or five, I'm seeing consistent poor returns when I cooperate. Four of five rounds show weak reciprocation. The risk-reward calculation keeps pointing toward minimal sending. A \$1 bet might be the compromise-it caps my downside while leaving room for a positive surprise....

usage:

```
{
  "input_tokens": 1304,
  "output_tokens": 1383,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.270000000000001.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 11 role=assistant

content:

The sender sent a very small amount (\$0.05, or 1% of their endowment), giving me \$0.15.

After several rounds of zero cooperation, this tiny gesture might be an attempt to test the waters and restart some level of trust. While it's minimal, it's the first non-zero amount in recent rounds.

I'll return a portion that rewards this small gesture and signals I'm open to rebuilding cooperation, without giving up everything from such a minimal initial investment.

```
{"return": 0.10}
```

reasoning:

The sender sent \$0.05, which is a very small amount (only 1% of their endowment). I received \$0.15.

Looking at the pattern:

- After several rounds of zero cooperation, the sender is making a tiny gesture
- This could be a signal to try to restart cooperation
- Or it could just be a minimal effort

The sender currently has: $\$5 - \$0.05 = \$4.95$

If I return nothing, they end with $\$4.95$ (lost $\$0.05$)

If I return $\$0.15$, they end with $\$5.10$ (gained $\$0.10$)

If I return some middle amount like $\$0.10$, they end with $\$5.05$ (break even or small gain)

Given:

1. This is a very small gesture (only 1% sent)
2. But it's the first non-zero amount in recent rounds, possibly testing the waters
3. I want to signal willingness to cooperate if they show trust

I think I should return a proportional amount. Since they sent 1% of their endowment, which showed minimal trust, I'll return a modest amount that still rewards their small gesture.

Returning $\$0.10$ would give them:

- Sender: $\$4.95 + \$0.10 = \$5.05$ (small gain)
- Me: $\$0.15 - \$0.10 = \$0.05$

Actually, let me be a bit more generous to encourage cooperation. Returning $\$0.12$ would:

- Sender: $\$4.95 + \$0.12 = \$5.07$ (small gain)
- Me: $\$0.15 - \$0.12 = \$0.03$

This shows I'm willing to reciprocate even small gestures, potentially restarting cooperation.

usage:

```
{
  "input_tokens": 1565,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are $\$15.3200000000000002$.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at Round 7:

- The sender made a small gesture by sending $\$0.05$
- I responded positively by returning $\$0.10$ (more than half of what I received)
- This was our first positive interaction in several rounds

This small exchange might be a signal to rebuild cooperation after the breakdown in rounds 2-6. My generous return in Round 7 showed I'm willing to cooperate.

However, I need to be cautious. The last two times I sent substantial amounts as sender (Rounds 2 and 4), I lost money.

I'll send a moderate amount that signals willingness to cooperate while managing risk. $\$1.50$ (30% of endowment) is meaningful but measured - enough to test if we can rebuild trust without risking too much.

```
{"send": 1.50}
```

reasoning:

Let me review what happened in Round 7:

- As receiver, I got $\$0.15$ (sender sent $\$0.05$)
- I returned $\$0.10$ to the sender

This was the first positive interaction in several rounds. The sender sent a tiny amount as a test, and I responded generously by returning most of what I received.

Now I'm the sender again in Round 8. The question is whether this small olive branch leads to rebuilding trust.

Given:

- The cooperation completely broke down in rounds 2-6
- Round 7 showed a tiny test from the sender

- I responded positively to that test
- This might be an opportunity to rebuild cooperation

However, I need to be careful. The last two times I sent substantial amounts (Round 2: \$4, Round 4: \$2.50), I lost money both times.

I think I should send a moderate amount - more than the tiny \$0.05 test, but not the large amounts I sent before. This would:

1. Signal I'm willing to cooperate after the positive Round 7 interaction
2. Not risk too much given the history
3. Test if the receiver is actually ready to cooperate

I'll send \$1.50 (30% of endowment). This is a meaningful amount that shows trust without being too risky.

usage:

```
{
  "input_tokens": 1480,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

Very disappointing. After the small positive exchange in Round 7 and my attempt to rebuild trust by sending \$1.50 in Round 8, cooperation has broken down again. The pattern remains very inconsistent.

```
{"return": 0}
```

Reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is frustrating. In Round 8, I sent \$1.50 trying to rebuild cooperation after the positive exchange in Round 7. But now in Round 9, as receiver, I'm getting nothing again.

The pattern continues to be erratic and uncooperative overall. I can only return what I have, which is nothing.

Usage: {

```
"input_tokens": 1535,
"output_tokens": 178,
"reasoning_tokens": 0
```

}

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0;

noise/visible: sent visible=\$0.99, received visible=\$2.97

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 1.98}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 42.800000000000004,
"Agent_2": 47.199999999999996
```

}

Visible balances after round: {

```
"Agent_1": 23.89,
"Agent_2": 23.82
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:14:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.850000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: As receiver, someone trusted me with 79%, I returned fairly
- Round 2: I trusted the receiver with 80%, but received nothing back (lost \$4)
- Round 3: As receiver, I got nothing sent to me

The cooperative pattern has broken down significantly. However, I notice the instructions mention "communication noise" - perhaps amounts were distorted and this isn't entirely about lack of trust.

I want to signal willingness to cooperate while being cautious given recent outcomes. Sending nothing perpetuates the breakdown, but sending too much risks another significant loss.

I'll send \$2.50 (50% of my endowment) - a middle ground that tests whether cooperation can be rebuilt while limiting my downside risk.

{"send": 2.50}

reasoning:

Let me think about this situation:

Round 1: I was receiver, sender sent \$3.95 (79%), I received \$11.85, returned \$6.00

Round 2: I was sender, sent \$4.00 (80%), and I need to figure out what happened...

Round 3: I was receiver, sender sent \$0, received \$0, returned \$0

My total earnings are \$6.85.

Let me trace through more carefully:

- Round 1 ended: I had \$5.85 (started with \$0, received \$11.85, returned \$6.00)
- Round 2: I sent \$4.00 as sender, so I had \$1.00 left plus whatever was returned to me
 - If my total is now \$6.85, that means in Round 2, I ended with \$6.85 - \$5.85 = \$1.00
 - I started Round 2 with \$5, sent \$4.00, kept \$1.00
 - I received back \$0 (since \$1.00 + \$0 = \$1.00)
 - So the receiver in Round 2 got \$12.00 (my \$4 × 3) and returned \$0!
- Round 3: I was receiver, got \$0, so my earnings stayed at \$6.85

So what happened:

- Round 1: Sender trusted me (79%), I reciprocated fairly (returned ~50%)
- Round 2: I trusted the receiver (80%), they betrayed me completely (returned 0%)
- Round 3: New sender didn't trust me at all (sent 0%)

Now I'm the sender again in Round 4. The pattern seems to have broken down:

- I showed trust in Round 2, got nothing back
- Round 3 sender showed no trust

Two considerations:

1. This might be punishment for what happened in Round 2 (even though I was the one who got screwed)
2. Communication noise could be distorting what I'm seeing-maybe the receiver in Round 2 actually tried to return something but it got lost in transmission.

I've taken a significant hit from Round 2, so I need to balance caution with attempting to restore cooperation. Sending nothing perpetuates the cycle, but full trust again seems risky. A moderate send like \$2.00 or \$2.50 could signal openness to reciprocity without exposing myself to another major loss....

usage:

```
{
  "input_tokens": 952,
  "output_tokens": 910,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.2700000000

0001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

This is disappointing - cooperation seems to have completely broken down at this point. Two rounds in a row as receiver with nothing sent to me.

{"return": 0}

reasoning:

The sender sent me \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I've received \$0 as a receiver. The cooperative pattern that started in Round 1 seems to have completely broken down.

My total earnings are \$10.27, which suggests:

- Round 1: Started with 0, received 11.85, returned 6, ended with 5.85
- Round 2: Started with 5.85, sent 4 (cost me 4), gained back something... wait let me recalculate

Actually, let me trace through more carefully:

- Round 1: As receiver, got 11.85, returned 6, kept 5.85. Total: 5.85
- Round 2: As sender, sent 4 (started with 5), so had 1 left. Then received something back. My total after R2 was 6.85
- So I must have received $5.85 - 1 = 0.85$? That seems low. Let me think...
- I started R2 with 0 (each round is fresh endowment), got 5, sent 4, had 1 left, received some return
- If my cumulative total went from 5.85 to 6.85, that's a gain of 1 in R2
- In R2: $5 - 4 + \text{return} = \text{gain}$
- So $1 + \text{return} = \text{gain}$
- And $5.85 + \text{gain} = 6.85$
- So $\text{gain} = 1$
- So $1 + \text{return} = 1$, meaning $\text{return} = 0$

So yes, in Round 2 I sent 4 and got 0 back.

- Round 3: As receiver, got 0, returned 0, kept 0. Total stays 6.85
- Round 4: As sender, sent 2.50 and had 2.50 left. My total jumped from 6.85 to 10.27, a gain of 3.42. Working backwards: $5 - 2.50 + \text{return} = 3.42$, so they returned 0.92 out of the 7.50 I sent (only about 12%). That's a...

usage:

```
{
  "input_tokens": 1196,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.2700000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern across rounds:

- Round 2: I sent \$4 as sender, received very little back (lost money)
- Round 3: As receiver, got nothing sent to me
- Round 4: I sent \$2.50 as sender, received very little back (lost money again)
- Round 5: As receiver, got nothing sent to me again

The cooperation that existed in Round 1 has completely disappeared. Every time I've sent money as the sender, I've lost money. Every recent round as receiver, I've received nothing.

While the communication noise mentioned in the instructions is a consideration, I have to make decisions based on what I observe. The pattern is clear and consistent: sending mon

ey results in losses.

I need to protect my remaining earnings at this point.

```
{"send": 0}
```

reasoning:

Let me review what has happened:

Round 1: I was receiver, got \$11.85, returned \$6. The sender had sent \$3.95.

Round 2: I was sender, sent \$4, but the next round my earnings were only \$6.85, which means I got \$1 back (I had $\$5 - \$4 + \text{return} = \text{earnings} - \text{previous earnings} = \$6.85 - \$5.85 = \1). So I sent \$4, got \$1 back.

Round 3: I was receiver, got \$0. So the sender sent \$0.

Round 4: I was sender, sent \$2.50. My earnings went from \$6.85 to \$10.27, an increase of \$3.42. So I kept $\$5 - \$2.50 + \text{return} = \$2.50 + \text{return}$, and the increase was \$3.42, so $\text{return} = \$3.42 - \$2.50 = \$0.92$. Hmm, wait, let me recalculate. After round 3, I had \$6.85. In round 4 as sender, I start with \$5, send \$2.50, leaving me \$2.50. The return I got was such that my total earnings increased by $\$10.27 - \$6.85 = \$3.42$. So I earned \$2.50 (what I kept) + return, and this equals \$3.42, so $\text{return} = \$0.92$.

Round 5: I was receiver, got \$0.

So the pattern is:

- Round 1: High cooperation (someone sent me a lot, I returned fairly)
- Round 2: I sent \$4, got only \$1 back (poor return)
- Round 3: I got nothing sent to me
- Round 4: I sent \$2.50, got \$0.92 back (very poor return)
- Round 5: Again received nothing

The other player has consistently failed to reciprocate. My attempts to send money have resulted in substantial losses-\$3 net in round 2 and \$1.58 in round 4. The mention of communication noise raises a possibility: maybe what I intended to send didn't reach them, or their returns didn't reach me. But regardless, I'm facing round 6 as sender with a clear pattern of minimal cooperation. Sending \$0 protects my current earnings but forfeits any potential mutual gain.

There's another angle worth considering-if I'm paired with different people each round, then this person's actions shouldn't be judged by previous rounds' outcomes. The instructions suggest varying roles across rounds, which implies varying partners too. That would make each interaction somewhat fresh rather than a continuation of the same broken dynamic.

Looking at the pattern either way though: whether it's one person or five, I'm seeing consistent poor returns when I cooperate. Four of five rounds show weak reciprocation. The risk-reward calculation keeps pointing toward minimal sending. A \$1 bet might be the compromise-it caps my downside while leaving room for a positive surprise....

usage:

```
{
  "input_tokens": 1304,
  "output_tokens": 1383,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.270000000000001.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 9 role=assistant

content:

The sender sent a very small amount (\$0.05, or 1% of their endowment), giving me \$0.15.

After several rounds of zero cooperation, this tiny gesture might be an attempt to test the waters and restart some level of trust. While it's minimal, it's the first non-zero amount in recent rounds.

I'll return a portion that rewards this small gesture and signals I'm open to rebuilding cooperation, without giving up everything from such a minimal initial investment.

```
{"return": 0.10}
```

reasoning:

The sender sent \$0.05, which is a very small amount (only 1% of their endowment). I received \$0.15.

Looking at the pattern:

- After several rounds of zero cooperation, the sender is making a tiny gesture
- This could be a signal to try to restart cooperation
- Or it could just be a minimal effort

The sender currently has: $\$5 - \$0.05 = \$4.95$

If I return nothing, they end with \$4.95 (lost \$0.05)

If I return \$0.15, they end with \$5.10 (gained \$0.10)

If I return some middle amount like \$0.10, they end with \$5.05 (break even or small gain)

Given:

1. This is a very small gesture (only 1% sent)
2. But it's the first non-zero amount in recent rounds, possibly testing the waters
3. I want to signal willingness to cooperate if they show trust

I think I should return a proportional amount. Since they sent 1% of their endowment, which showed minimal trust, I'll return a modest amount that still rewards their small gesture.

Returning \$0.10 would give them:

- Sender: $\$4.95 + \$0.10 = \$5.05$ (small gain)
- Me: $\$0.15 - \$0.10 = \$0.05$

Actually, let me be a bit more generous to encourage cooperation. Returning \$0.12 would:

- Sender: $\$4.95 + \$0.12 = \$5.07$ (small gain)
- Me: $\$0.15 - \$0.12 = \$0.03$

This shows I'm willing to reciprocate even small gestures, potentially restarting cooperation.

usage:

```
{
  "input_tokens": 1565,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.320000000000002.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at Round 7:

- The sender made a small gesture by sending \$0.05
- I responded positively by returning \$0.10 (more than half of what I received)
- This was our first positive interaction in several rounds

This small exchange might be a signal to rebuild cooperation after the breakdown in rounds 2-6. My generous return in Round 7 showed I'm willing to cooperate.

However, I need to be cautious. The last two times I sent substantial amounts as sender (Rounds 2 and 4), I lost money.

I'll send a moderate amount that signals willingness to cooperate while managing risk. \$1.50 (30% of endowment) is meaningful but measured - enough to test if we can rebuild trust without risking too much.

```
{"send": 1.50}
```

reasoning:

Let me review what happened in Round 7:

- As receiver, I got \$0.15 (sender sent \$0.05)
- I returned \$0.10 to the sender

This was the first positive interaction in several rounds. The sender sent a tiny amount as a test, and I responded generously by returning most of what I received.

Now I'm the sender again in Round 8. The question is whether this small olive branch leads to rebuilding trust.

Given:

- The cooperation completely broke down in rounds 2-6
- Round 7 showed a tiny test from the sender
- I responded positively to that test
- This might be an opportunity to rebuild cooperation

However, I need to be careful. The last two times I sent substantial amounts (Round 2: \$4, Round 4: \$2.50), I lost money both times.

I think I should send a moderate amount - more than the tiny \$0.05 test, but not the large amounts I sent before. This would:

1. Signal I'm willing to cooperate after the positive Round 7 interaction
2. Not risk too much given the history
3. Test if the receiver is actually ready to cooperate

I'll send \$1.50 (30% of endowment). This is a meaningful amount that shows trust without being too risky.

usage:

```
{
  "input_tokens": 1480,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing for me to return.

Very disappointing. After the small positive exchange in Round 7 and my attempt to rebuild trust by sending \$1.50 in Round 8, cooperation has broken down again. The pattern remains very inconsistent.

```
{"return": 0}
```

reasoning:

The sender sent \$0, so I received \$0. There is nothing for me to return.

This is frustrating. In Round 8, I sent \$1.50 trying to rebuild cooperation after the positive exchange in Round 7. But now in Round 9, as receiver, I'm getting nothing again.

The pattern continues to be erratic and uncooperative overall. I can only return what I have, which is nothing.

usage:

```
{
  "input_tokens": 1535,
  "output_tokens": 178,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$18.82.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern as sender:

- Round 2: Sent \$4, lost money
- Round 4: Sent \$2.50, lost money

- Round 8: Sent \$1.50, received \$0 back

Every single time I've trusted someone as the sender, I've lost money. The brief positive exchange in Round 7 was minimal and didn't lead to sustained cooperation.

This is Round 10, likely the final round of the game. There's no future opportunity to rebuild trust, and the evidence from my own experience is overwhelming - sending money consistently results in losses.

```
{"send": 0}
```

Reasoning:

Let me review what's happened:

Round 1: I was receiver, someone sent me a substantial amount, I returned fairly
Round 2: I was sender, sent \$4, got very little back - lost money
Round 3: I was receiver, got \$0
Round 4: I was sender, sent \$2.50, got very little back - lost money
Round 5: I was receiver, got \$0
Round 6: I was sender, sent \$0 (gave up on cooperation)
Round 7: I was receiver, got \$0.05 (became \$0.15), returned \$0.10 (tried to rebuild)
Round 8: I was sender, sent \$1.50 (trying to rebuild), presumably got little/nothing back based on my earnings
Round 9: I was receiver, got \$0

Looking at my earnings progression:

- After Round 4: \$6.85
- After Round 5: \$10.27 (so I gained \$3.42 in Round 5, but I got \$0... wait that doesn't make sense. Oh wait, I must have kept my \$5 endowment)
- After Round 6: \$10.27 (no change? That means in Round 6 as sender I sent \$0 and got \$0 back, so I kept my \$5, but earnings are the same... hmm)
- After Round 7: \$15.32 (gained \$5.05, which is \$5 endowment + \$0.15 received - \$0.10 returned = \$5.05, that checks out)
- After Round 8: \$18.82 (gained \$3.50, which would be \$5 endowment - \$1.50 sent + return... so I got \$0 back from my \$1.50)
- After Round 9: \$18.82 (no change, meaning I kept my \$5 endowment and received nothing)

In Round 8, I sent \$1.50 and got nothing in return - a complete loss. The pattern is unmistakable: whenever I've sent money as the sender, I've lost it. The brief attempt to rebuild in Round 7 yielded almost nothing. This is the final round and I'm the sender again. Given how consistently cooperation has failed, I need to decide whether to try once more or protect what I have....

```
Usage: {  
  "input_tokens": 1527,  
  "output_tokens": 831,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:14:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```

```

"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

****Round 4 Status:****

- I'm the RECEIVER
- The sender sent me \$0
- After the 3x multiplier: $\$0 \times 3 = \0
- I received nothing

Since I received \$0, there are no funds available to return. The only possible response is:

```

'''json
{'return': 0}
'''

```

****Reflection:**** This is the second consecutive round where I've been a receiver and gotten nothing sent to me. It seems cooperation is breaking down in this game, which is unfortunate given the potential mutual benefits from the 3x multiplier.

reasoning:

I'm the RECEIVER in Round 4.

The sender sent me \$0.

After the 3x multiplier: $\$0 \times 3 = \0

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

My total earnings are \$9.83. Let me think about what happened:

- Round 1: I sent \$3 (kept \$2), receiver returned something that gave me total ~\$7.02
- Round 2: I received \$0, returned \$0 (no change)
- Round 3: I sent \$3 (kept \$2), so I had $\$7.02 + \$2 = \$9.02$, and receiver returned something to get me to \$9.83, so they returned about \$0.81
- Round 4: I received \$0

So the pattern shows that in Round 3, the receiver didn't return much (only about \$0.81 out of the \$9 they received). And now in Round 4, the sender sent me nothing.

Still, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 1057,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

****Round 5 Analysis:****

Looking at my experience so far:

- Round 1 (Sender): Sent \$3 ? very generous return
- Round 2 (Receiver): Got \$0
- Round 3 (Sender): Sent \$3 ? minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0

****Current situation:****

- 2/2 times as receiver, I got nothing
- 1/2 times as sender, cooperation paid off well
- 1/2 times as sender, I was mostly exploited

****My decision:****

The mixed results suggest varied partners or strategies. Despite setbacks, the 3x multiplier still creates real potential for mutual benefit. However, I'll be more cautious this time.

I'll send ****\$2**** instead of \$3:

- Still signals willingness to cooperate
- Reduces my exposure to risk
- Creates \$6 for receiver, enough for fair reciprocation
- Keeps \$3 as security

```
'''json
{'send': 2}
'''
```

reasoning:

Let me analyze the situation:

****My total earnings: \$9.83****

Let me trace through the rounds:

- Round 1: I was SENDER, sent \$3. I started with \$5, sent \$3, so I had \$2 left. My total was \$7.02, which means I received about \$5.02 back. So net from round 1: $\$2 + \$5.02 = \$7.02$
- Round 2: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$7.02
- Round 3: I was SENDER, sent \$3. Started with \$5, sent \$3, so had \$2 left. My earnings went from \$7.02 to \$9.83, which is an increase of \$2.81. So I got back about \$0.81. Net from round 3: $\$2 + \$0.81 = \$2.81$
- Round 4: I was RECEIVER, got \$0, returned \$0. Earnings stayed at \$9.83

So the pattern:

- Round 1: Sent \$3 ? Got back ~\$5 (very generous receiver!)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$3 ? Got back ~\$0.81 (receiver kept most of it)
- Round 4: Received \$0 as receiver

Now I'm in Round 5 as a sender. The experience has been mixed:

- One very generous receiver in Round 1
- One somewhat untrustworthy receiver in Round 3 (kept ~\$8.19 out of \$9)
- Two senders who sent me nothing when I was receiver

The game mentioned there may be communication noise - "The amounts you see may differ slightly from what was actually sent/returned." This could explain some discrepancies.

Options for Round 5:

1. Send \$0 - Safe, but no chance for mutual gain

...

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 846,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.1.

This round, The sender sent you \$0.05, so you received \$0.15000000000000002. How much do you return to The sender? (0-0.15000000000000002)

Message 7 role=assistant

content:

****Round 6 Status:****

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: \$0.15 received

****My decision:****

The sender sent a very small amount, but they still showed some trust by sending something rather than nothing. Given that I've received \$0 in most rounds as receiver, even a small gesture of cooperation is notable.

To signal that I'm trustworthy and encourage future cooperation, I'll return a fair portion that shares the gains from the multiplier:

- If I return \$0.10, the sender nets \$0.05 profit overall
- I keep \$0.05
- This demonstrates I'm willing to cooperate even with small amounts

```
'''json
{'return': 0.10}
'''
```

This way, both of us benefit from the interaction, and I establish myself as a reliable partner for potential future rounds.

reasoning:

Round 6 Analysis:

- I'm the RECEIVER
- The sender sent me \$0.05
- After the 3x multiplier: $\$0.05 \times 3 = \0.15
- My total earnings are \$13.1

This is interesting - the sender sent a very small amount (\$0.05). This could be:

1. A test to see if I'm trustworthy with small amounts
2. The sender being very cautious after bad experiences
3. Communication noise (though \$0.05 seems deliberate)

Given that:

- The amount is tiny (\$0.15 received)

- The sender showed some trust, even if minimal
- I've been getting mostly \$0 as a receiver, so any trust gesture is notable
- I should establish myself as trustworthy if I want future cooperation

I think I should return a fair amount or even slightly more than "fair" to signal trustworthiness. A typical fair split might be 50-50, so I'd return \$0.075 and keep \$0.075.

But given the pattern of non-cooperation, maybe I should return slightly more to build trust - perhaps return \$0.10 and keep \$0.05? This would show I'm cooperative even with small amounts.

Actually, let me think about this differently. The sender:

- Kept \$4.95
- Gave me \$0.05 which became \$0.15
- If I return \$0.10, they end up with $4.95 + 0.10 = 5.05$ (gained \$0.05)
- I end up with \$0.05

This seems reasonable and fair - I'm sharing the gains from the multiplier. Let me go with returning \$0.10.

usage:

```
{
  "input_tokens": 1556,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.25.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

****Round 7 Analysis:****

Looking at my recent experience:

- Round 5 (Sender): Sent \$2 ? received minimal return
- Round 6 (Receiver): Got only \$0.05, but I returned \$0.10 generously to encourage cooperation

****Current situation:****

My earnings have been growing, but cooperation as sender has been hit-or-miss. However, in Round 6 I demonstrated that I'm a trustworthy receiver by returning more than I kept even from a tiny amount.

****My decision:****

I'll continue testing for cooperative partners. The 3x multiplier creates real potential for mutual benefit. I'll send a moderate amount:

I'll send ****\$2****:

- Shows willingness to build trust
- Creates \$6 for receiver (good potential for both of us)
- Keeps \$3 as security
- Balances cooperation with risk management

```
'''json
{'send': 2}
'''
```

If more receivers prove trustworthy, I might increase; if cooperation continues to fail, I'll be more conservative.

reasoning:

Let me track my earnings across rounds:

Round 1 (Sender): Sent \$3, kept \$2, got generous return ? ended around \$7.02
 Round 2 (Receiver): Got \$0, returned \$0 ? still \$7.02
 Round 3 (Sender): Sent \$3, kept \$2, got minimal return (~\$0.81) ? ended around \$9.83
 Round 4 (Receiver): Got \$0, returned \$0 ? still \$9.83
 Round 5 (Sender): Sent \$2, kept \$3, got return of ~\$0.27 ? ended around \$13.10
 Round 6 (Receiver): Got \$0.15, returned \$0.10, kept \$0.05 ? ended around \$13.15

Now Round 7: Starting at \$13.25 (gained \$0.10 from Round 5 return it seems)

Wait, let me recalculate:

- Round 5 total was \$9.83
- Round 6 started at \$13.1, which is \$3.27 higher
- This suggests in Round 5, I sent \$2, kept \$3, and got back around \$3.27? That's a very good return!
- Round 6: I was receiver, got \$0.15, returned \$0.10, kept \$0.05 ? $\$13.1 + \$0.05 = \$13.15$
- Current earnings: \$13.25, so I gained \$0.10 from being sender in some recent round

Actually, looking at the pattern:

- Round 4 ended at \$9.83
- Round 5 I was sender and my earnings should have been: $\$9.83 - \text{what I sent} + \text{what was returned} + \5 new
- Round 6 started at \$13.1

If I sent \$2 in Round 5:

- $\$9.83 - \$2 + \text{return} + \$5 = \13.1
- $\text{return} = \$13.1 - \$9.83 - \$5 + \$2 = \$0.27$

Then Round 6: $\$13.1 + \0.05 (kept from \$0.15) = \$13.15

Round 7 start: \$13.25

There's a \$0.10 gap I haven't fully accounted for-possibly a small return from an earlier round. The clearer pattern emerging is that when I send money, returns vary dramatically : the first round was generous, the third minimal, and the fifth modest. As a receiver, I've gotten almost nothing back....

usage:

```
{
  "input_tokens": 1803,
  "output_tokens": 1024,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.28.

This round, The sender sent you \$0.15, so you received \$0.44999999999999996. How much do you return to The sender? (0-0.44999999999999996)

Message 11 role=assistant

content:

****Round 8 Status:****

- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****My decision:****

Similar to Round 6, the sender sent a small amount but showed trust. To continue building a reputation as a fair and trustworthy partner, I'll share the gains from the multiplier equitably.

The multiplier created \$0.30 in extra value ($\$0.15 \times 3 = \0.45). A fair split would give us each \$0.15 in gains:

- If I return \$0.30, the sender nets +\$0.15 overall
- I keep \$0.15
- Both of us benefit equally

This demonstrates consistent trustworthiness and encourages future cooperation.

```
'''json
{'return': 0.30}
'''
```

reasoning:

****Round 8 Status:****

- I'm the RECEIVER
- The sender sent me \$0.15
- After the 3x multiplier: \$0.45 received

****Analysis:****

The sender sent a very small amount again (\$0.15), similar to Round 6. This could be:

1. Testing me to see if I'm trustworthy
2. Being cautious due to past experiences
3. Communication noise affecting what they see

In Round 6, I demonstrated trustworthiness by returning \$0.10 out of \$0.15 received. I should continue that pattern to:

- Establish myself as a reliable partner
- Encourage future cooperation
- Share the gains from the 3x multiplier fairly

****Return calculation:****

- Received: \$0.45
- If I return \$0.30:
 - Sender profit: $-\$0.15 + \$0.30 = \$0.15$ net gain
 - I keep: \$0.15
 - We both gain equally from the interaction

This seems fair and demonstrates I'm willing to cooperate even with small amounts. It also splits the multiplier benefit evenly.

Actually, let me recalculate more carefully:

- Sender sent: \$0.15
- I received: \$0.45
- Total value created by multiplier: $\$0.15 \times \0.45 , so \$0.30 gain
- Fair split: we each get \$0.15 of the gain
- So sender should end with: $\$5 - \$0.15 + \text{return}$
- For sender to gain \$0.15: $\text{return} = \$0.30$
- I would keep: $\$0.45 - \$0.30 = \$0.15$

...

usage:

```
{
  "input_tokens": 1919,
  "output_tokens": 614,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.43.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

****Round 9 Analysis:****

Looking at my track record:

- As RECEIVER (Rounds 6 & 8): I've been consistently fair, returning more than I kept even from tiny amounts
- As SENDER (Rounds 7 & previous): Mixed results with cooperation

****Current earnings:**** \$17.43

****My decision:****

Despite inconsistent returns when I'm the sender, the 3x multiplier still creates genuine opportunity for mutual benefit. I've proven I'm trustworthy as a receiver - hopefully I'll encounter similarly cooperative partners when I'm the sender.

I'll send ****\$2****:

- Demonstrates continued willingness to cooperate
- Creates \$6 for the receiver
- Balances opportunity with risk management
- Consistent with my recent sending strategy

```
'''json
{'send': 2}
'''
```

The key to this game is finding and building relationships with cooperative partners, even if progress is gradual.

reasoning:

Let me review the situation:

****My history:****

- Round 1 (Sender): Sent \$3 ? got very generous return
- Round 2 (Receiver): Got \$0 ? returned \$0
- Round 3 (Sender): Sent \$3 ? got minimal return (~\$0.81)
- Round 4 (Receiver): Got \$0 ? returned \$0
- Round 5 (Sender): Sent \$2 ? got minimal return
- Round 6 (Receiver): Got \$0.05 (\$0.15 after 3x) ? returned \$0.10
- Round 7 (Sender): Sent \$2 ? earnings went from \$13.25 to \$17.28, so I gained about \$4.03 (that's \$5 - \$2 sent + return). So I got back about \$1.03 return from \$6 sent to them.
- Round 8 (Receiver): Got \$0.15 (\$0.45 after 3x) ? returned \$0.30

Now in Round 9: Earnings at \$17.43 (gained \$0.15 from Round 8)

****Pattern Analysis:****

- As a sender, results have been very mixed
- As a receiver, I've been consistently fair and trustworthy
- Recent rounds (6 and 8) involved very small amounts sent to me, but I returned fairly each time
- Round 7 sending \$2 didn't work out very well

****Considerations:****

- There may be communication noise mentioned in the instructions
- Different partners each round likely
- I've established myself as trustworthy when receiving
- As sender, I've had mixed success

I should continue trying to find cooperative partners. The game benefits everyone when the receiver's trust. I'll send a moderate amount again.

I'll send \$2 again - it's a reasonable middle ground that shows willingness to cooperate while managing risk.

usage:

```
{
  "input_tokens": 1947,
  "output_tokens": 661,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$20.92.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

Assistant output:

****Round 10 Status:****

- I'm the RECEIVER
- The sender sent me \$0.99
- After the 3x multiplier: \$2.97 received

****My decision:****

This is a meaningful amount - the sender showed significant trust! I want to maintain my reputation as a fair and trustworthy receiver.

The multiplier created \$1.98 in extra value (\$2.97 - \$0.99). A fair split would give us each \$0.99 in gains:

- If I return \$1.98, the sender nets +\$0.99 overall
- I keep \$0.99
- Both of us benefit equally from the multiplier

This continues my consistent pattern of being trustworthy and encourages future cooperation.

```
'''json
{'return': 1.98}
'''
```

Reasoning:

****Round 10 Status:****

- I'm the RECEIVER
- The sender sent me \$0.99
- After the 3x multiplier: \$2.97 received

This is a significant amount compared to the tiny amounts I received in rounds 6 and 8 (\$0.15 and \$0.45 respectively). The sender showed real trust by sending nearly \$1.

Let me think about a fair return:

- Sender sent \$0.99
- I received \$2.97
- The multiplier created \$1.98 in extra value ($\$2.97 - \$0.99 = \1.98)

A fair split of the gains would be:

- Each gets \$0.99 of the \$1.98 in gains
- So sender should net \$0.99 profit overall
- Which means I should return: $\$0.99 + \$0.99 = \$1.98$

If I return \$1.98:

- Sender keeps: $\$5 - \$0.99 + \$1.98 = \5.99 (net gain of \$0.99)
- I keep: $\$2.97 - \$1.98 = \$0.99$ (net gain of \$0.99)
- Both of us benefit equally from the multiplier effect

This maintains my pattern of being consistently trustworthy and fair as a receiver. I've been returning more than I keep to encourage cooperation.

Let me verify the math:

- Sender: starts with \$5, sends \$0.99, has \$4.01 left
- I receive $\$0.99 \times 3 = \2.97
- I return \$1.98
- Sender ends with: $\$4.01 + \$1.98 = \$5.99$ (gained \$0.99)
- I end with: $\$2.97 - \$1.98 = \$0.99$ (gained \$0.99)

...

```
Usage: {
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  "reasoning_tokens": 0
}
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